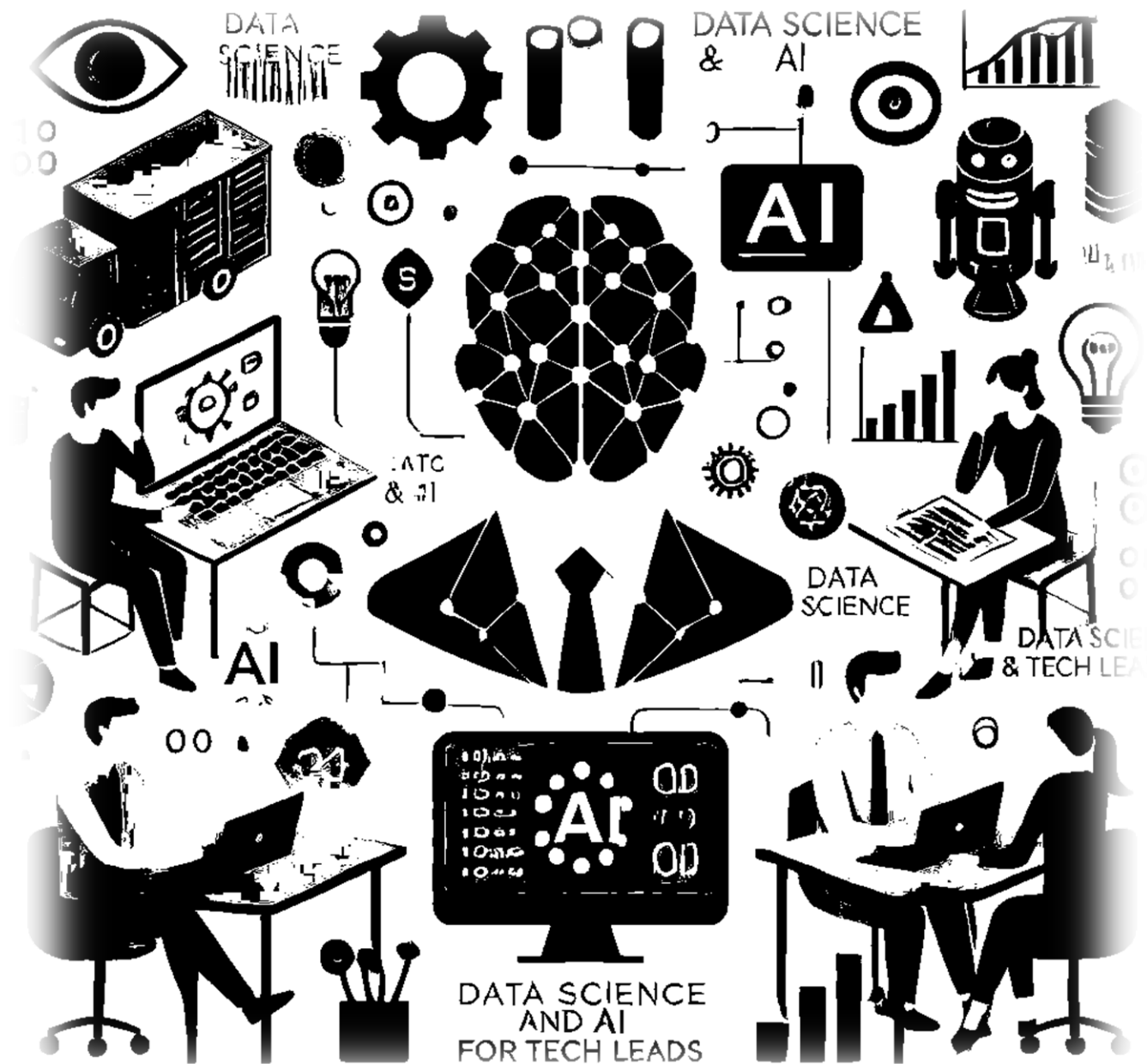


Especialização Tech Lead

Data Science & IA

2025.01

Prof. Vitor Casadei – vc@cesar.school



Agenda

1. Introdução
2. Aprendizado
3. Regressão Linear
4. Regressão Logística
5. Overfitting e Underfitting
6. Redes Neurais
7. Machine Learning e Deep Learning

Definição de Machine Learning

Machine learning: Field of study that **gives computers the ability to learn** without being explicitly programmed.

Arthur Samuel, 1959

Definição de Machine Learning

Well posed Learning Program: A computer is said to learn from experience **E** with respect to some class of task **T** and performance measure **P**, if it's performance at tasks in **T**, as measured by **P**, improves with experience **E**.

Tom Mitchell, 1998

Definição de Machine Learning

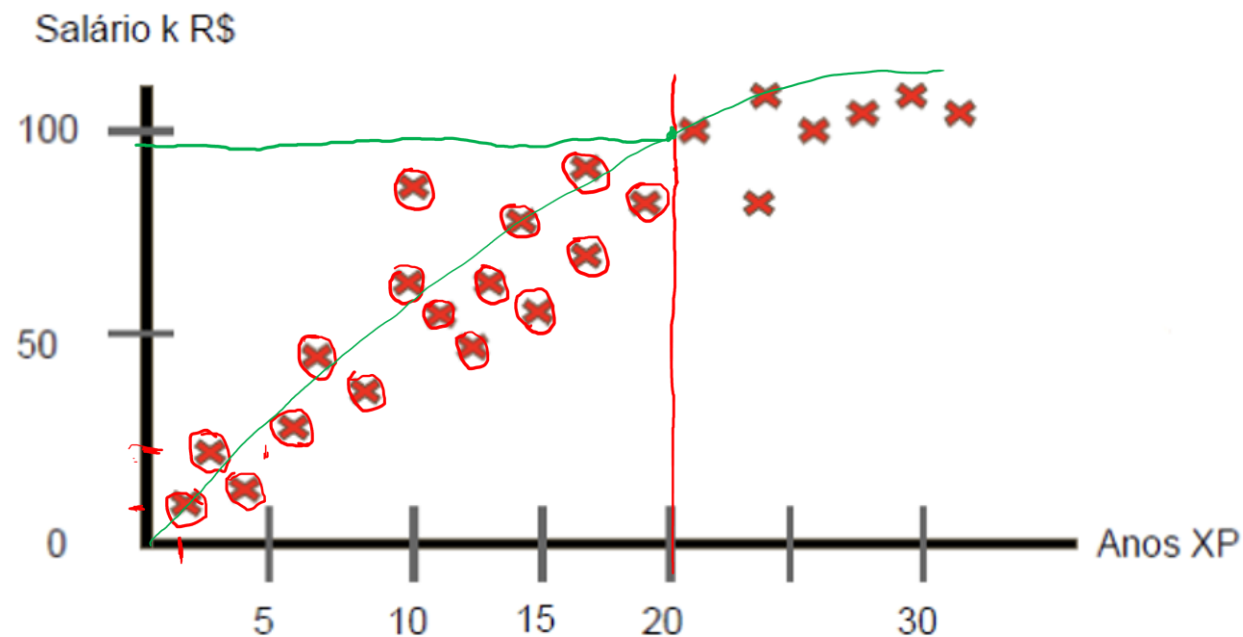
Well posed Learning Program: A computer is said to learn from experience **E** with respect to some class of task **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience **E**.

Tom Mitchell, 1998

Spam Classification:

- **E**: Watching you label emails as SPAM/Not SPAM
- **T**: Classifying emails as SPAM/Not SPAM
- **P**: The number (fraction) of emails correctly classified

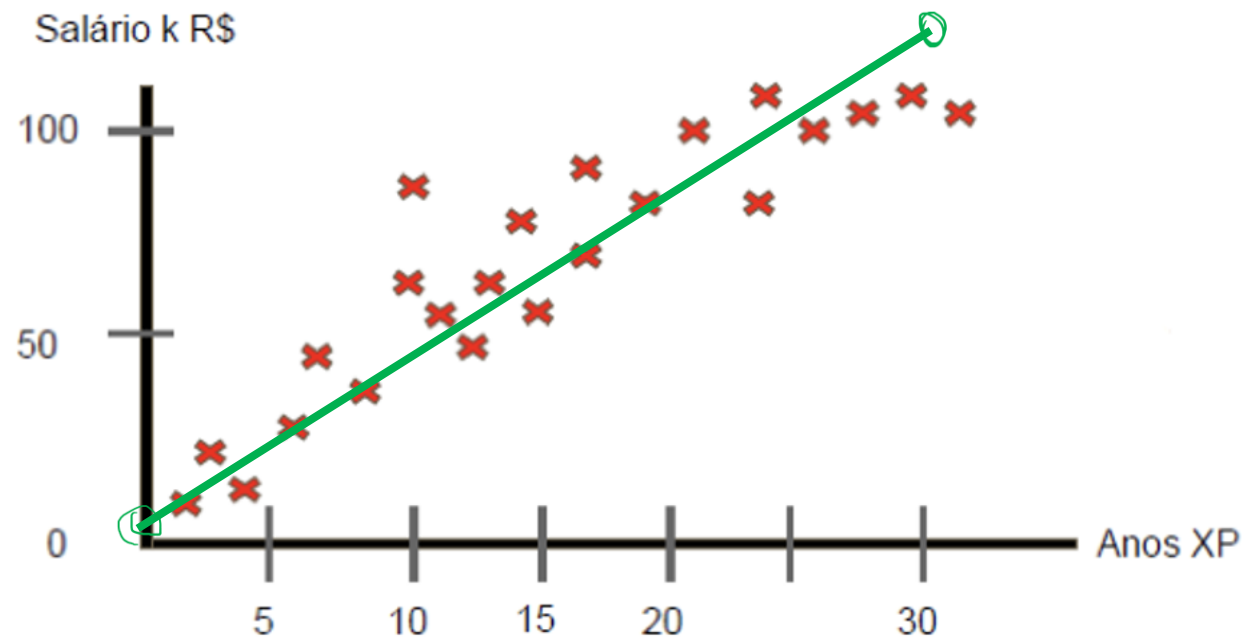
Aprendizado Supervisionado



Aprendizado com base nas
"respostas certas"

Regressão: Predição de Valores Contínuos

Aprendizado Supervisionado

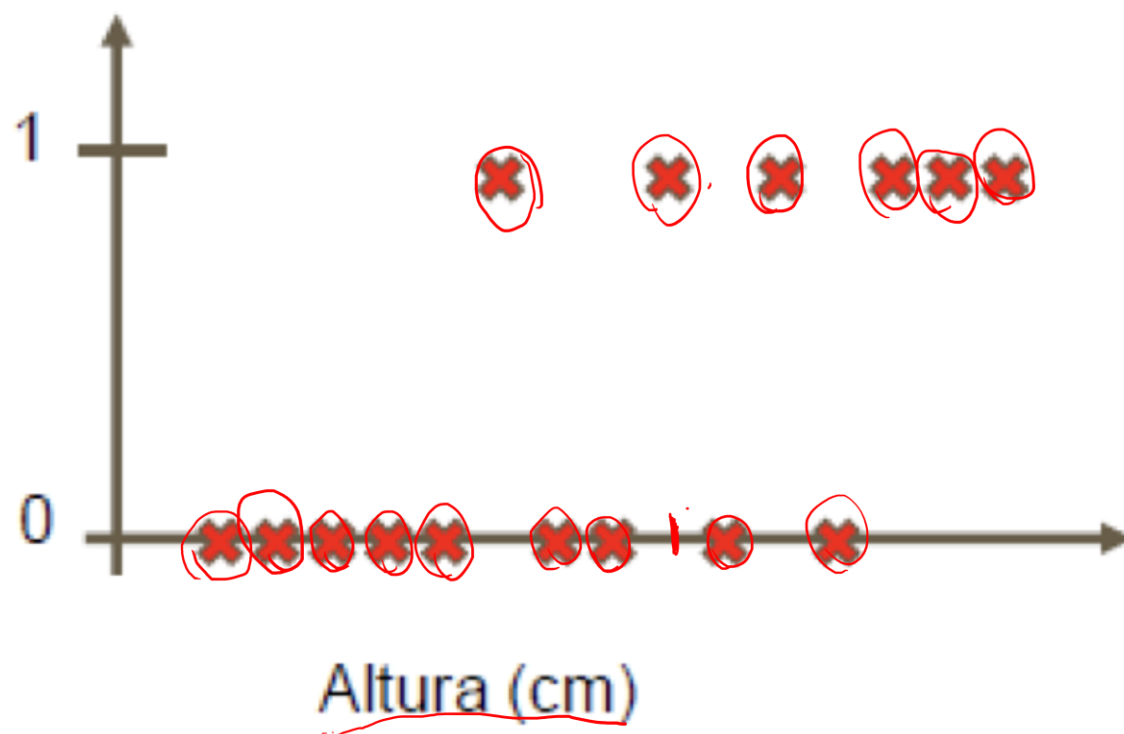


Aprendizado com base nas
"respostas certas"

Regressão: Predição de Valores Contínuos

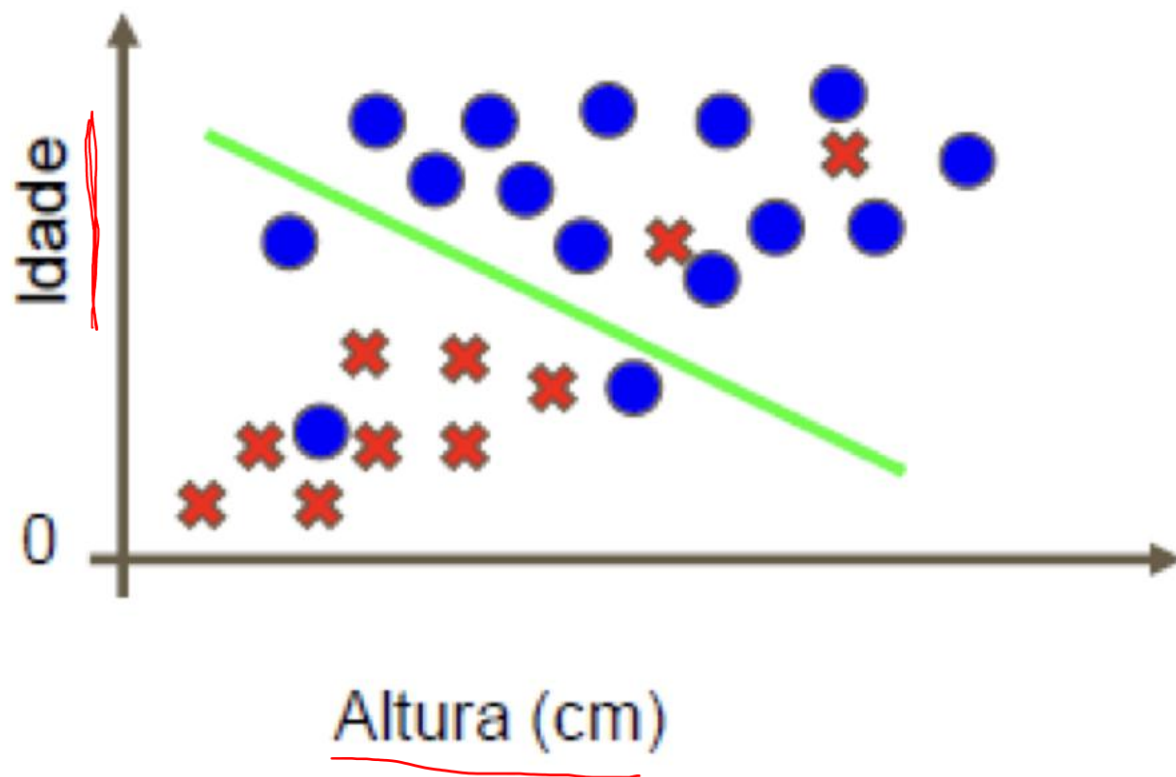
Aprendizado Supervisionado

Pessoa Adulto



Aprendizado Supervisionado

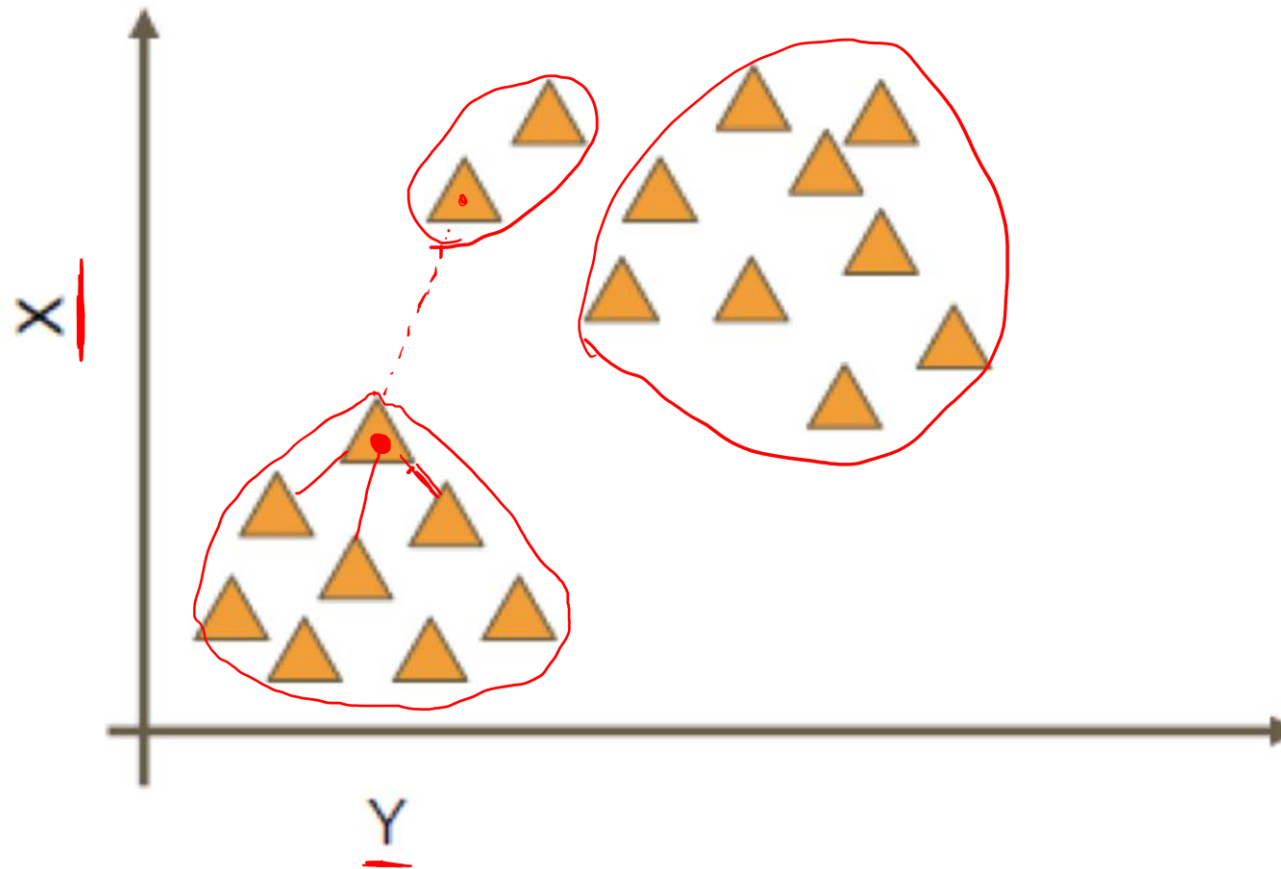
Pessoa Adulto



Aprendizado com base nas
"respostas certas"

Classificação: Predição de Valores Discretos

Aprendizado Não Supervisionado



Aprendizado Não Supervisionado

Interaction with mobile devices by elderly people: the Brazilian scenario.
Leme, R.R., Zaina, L.A. and Casadei, V., 2014



Manoel

Retired man, 70 years.

"Technology is not just for the younger"

Has a college degree, is an active member of Third Age group and well respected by people.
When friends of Mr. Manoel has doubts about new technologies, he readily help because they know he can not live without your smartphone, with which ships with the latest news.

Personal goals:

Do not waste time with technology and enjoy it the best way possible.

Practical goals:

The text of the programs on the mobile device could be a little bigger.



Marina

Retired woman, 80 years.

"I never had a cell phone!"

Although they enjoy good health, due to the limitations imposed by age, Marina depends on the family to perform basic tasks such as call to the grandchildren or watch a DVD. Regrets are just not having completed primary education. Carefree, proud to say I never laid a hand on a computer. Despite the insistence of the children, is reluctant to have a mobile device by consider it impossible to use.

Personal goals:

Enjoy life with their grandchildren. If you need the technology they can help me.

Practical goals:

The devices are too complicated to operate. I'm afraid to use them.



Lucia

A worker retired, 60 years.

"I'm retired but I can't stay without working"

Lucia is a person who although already retired, as the value of benefit is not enough, continues to work.
A few months ago bought a mobile device and is asking all friends how to use it. Regularly attend the computer classes and is finalizing the school. Is becoming a fan of games on mobile.

Personal goals:

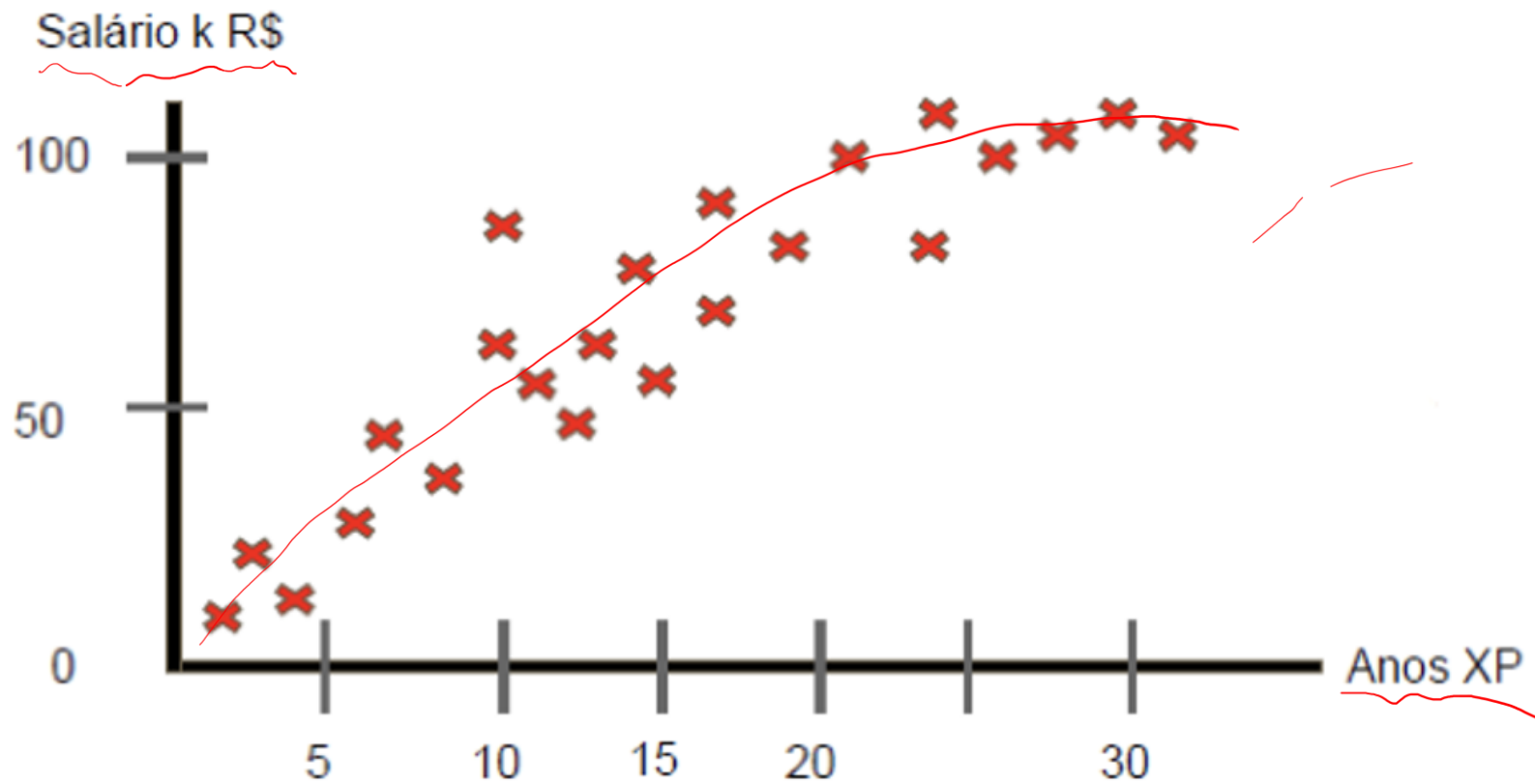
Learn more. Technology should support us constantly.

Practical goals:

At times I have trouble clicking on buttons, especially when the images are very small.

Regressão Linear – Uma variável

Predição de Salário Anual



Regressão Linear – Uma variável

Notação:

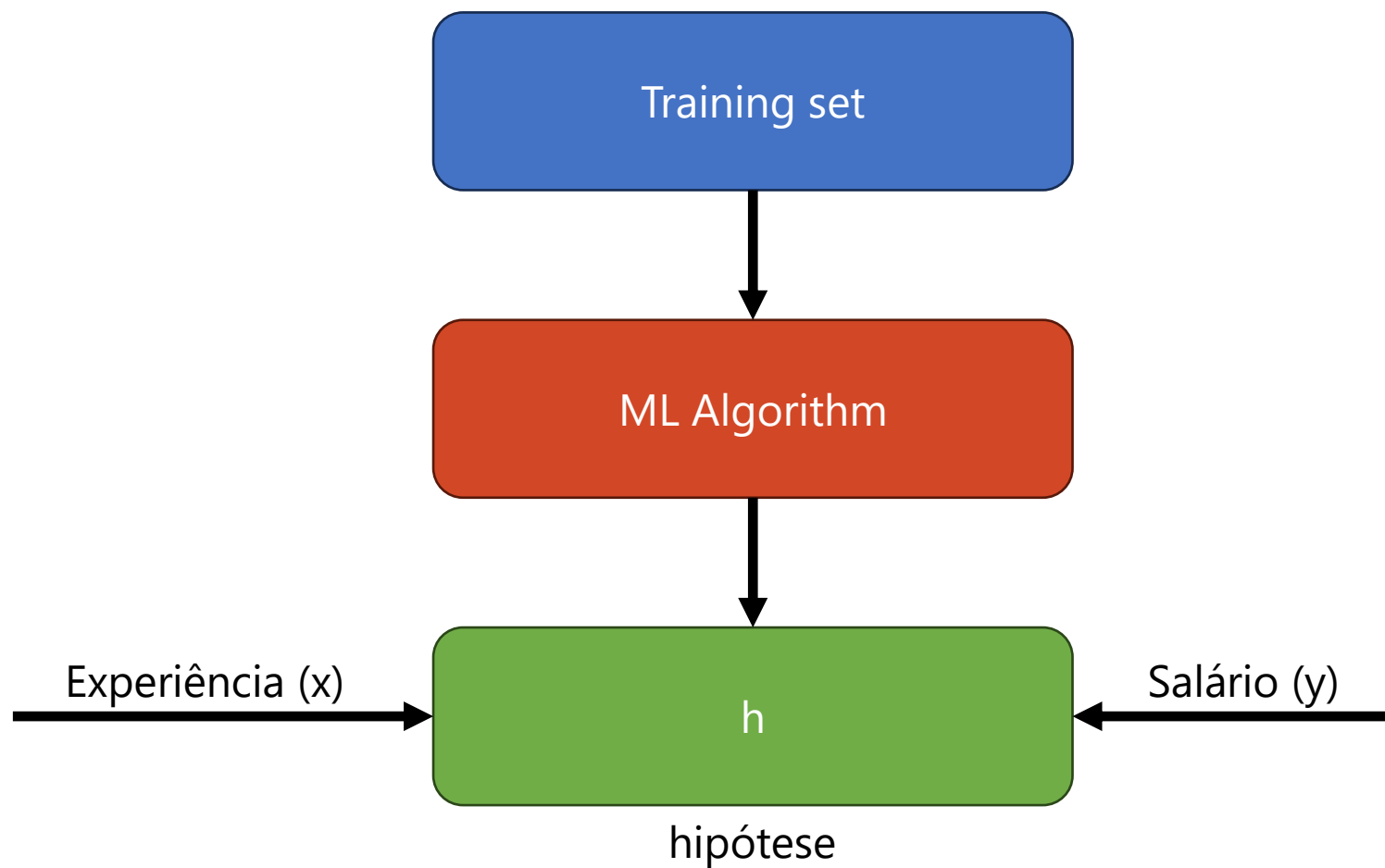
- **m**: número de exemplos de treinamento
- **x**: número de features
- **y**: target

Experiência (anos)	Salário Atual (K R\$)
2	36
8	50
13	72
20	98
...	...

x

y

Regressão Linear – Treinamento



Regressão Linear – Função de Custo

Notação:

- **m**: número de exemplos de treinamento
- **x**: features
- **y**: target

$$Y = \theta_0 + \theta_1 X$$

Experiência (anos)	Salário Atual (K R\$)
2	36
8	50
13	72
20	98
...	...

$\hat{y} = 15$

$\hat{y}$ $\hat{y}$

Hipótese: $\hat{y} = \theta_0 + \theta_1 \cdot x$

$\hat{y}, \text{pred}$

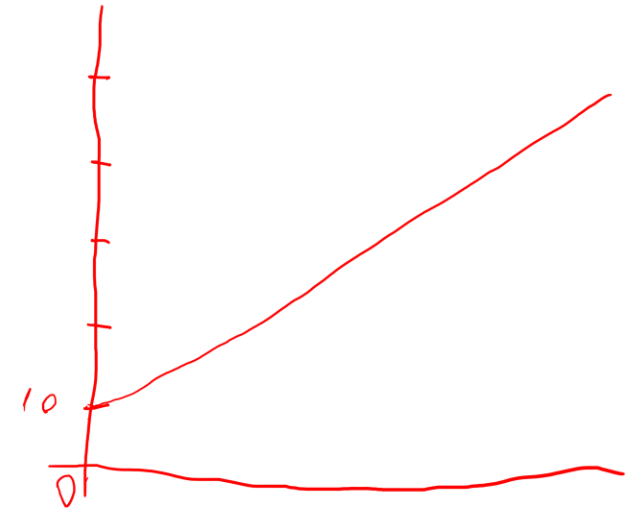
Regressão Linear – Função de Custo

Hipótese: $h(x) = \theta_0 + \theta_1 \cdot x$



$$h(x) = 40_0 + 0_1 \cdot x$$

$$\theta_0 = 40 \mid \theta_1 = 0$$

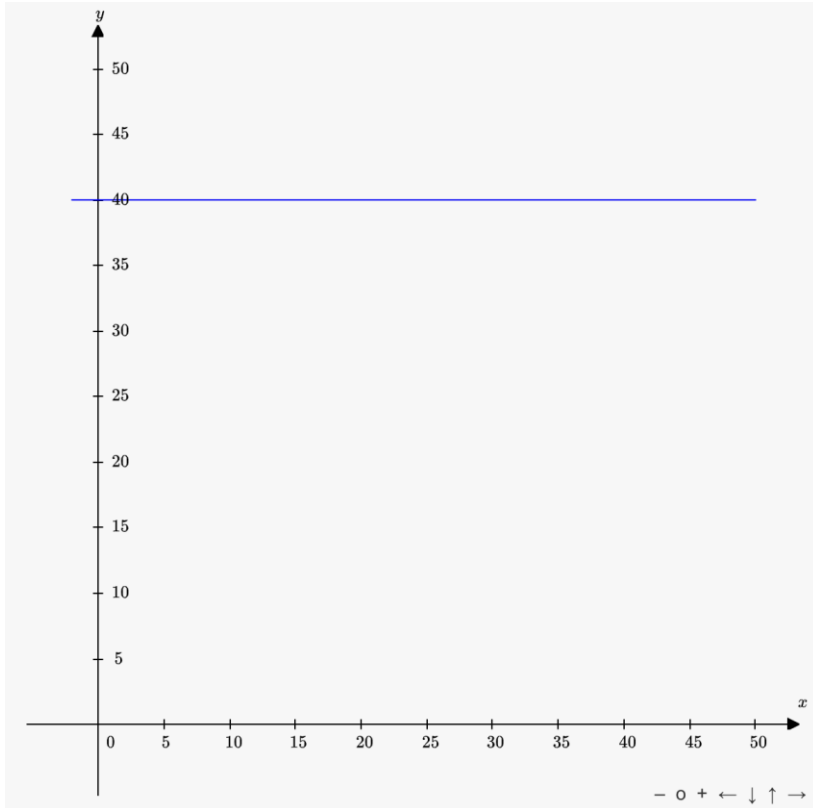


$$h(x) = 10_0 + 0.5_1 \cdot x$$

$$\theta_0 = 10 \mid \theta_1 = 0.5$$

Regressão Linear – Função de Custo

Hipótese: $h(x) = \theta_0 + \theta_1 \cdot x$



$$h(x) = 40_0 + 0_1 \cdot x$$

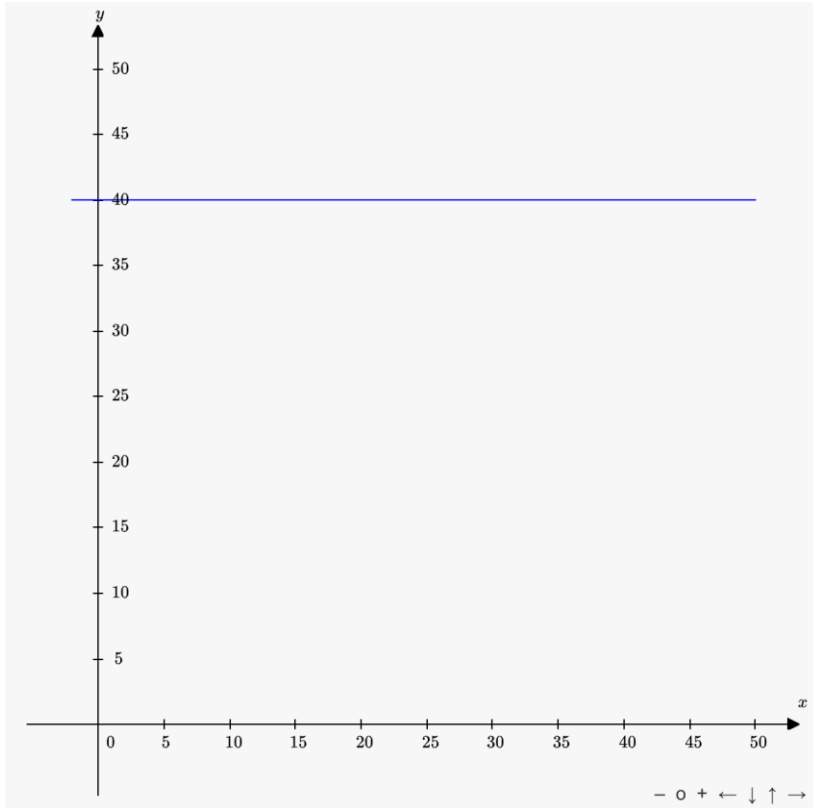
$$\theta_0 = 40 \mid \theta_1 = 0$$

$$h(x) = 10_0 + 0.5_1 \cdot x$$

$$\theta_0 = 10 \mid \theta_1 = 0.5$$

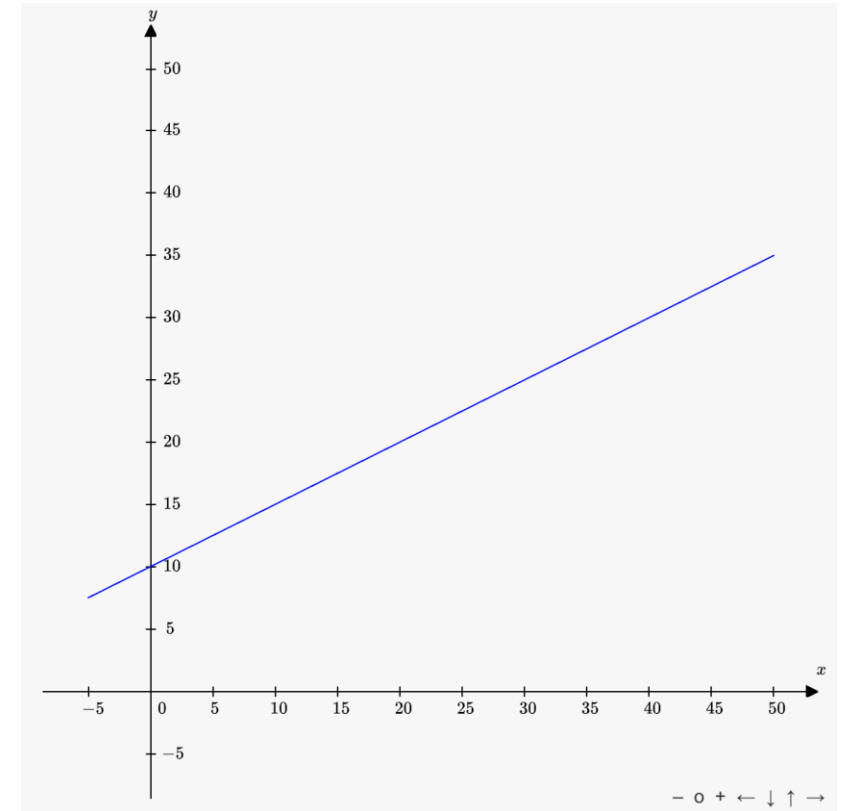
Regressão Linear – Função de Custo

Hipótese: $h(x) = \theta_0 + \theta_1 \cdot x$



$$h(x) = 40_0 + 0_1 \cdot x$$

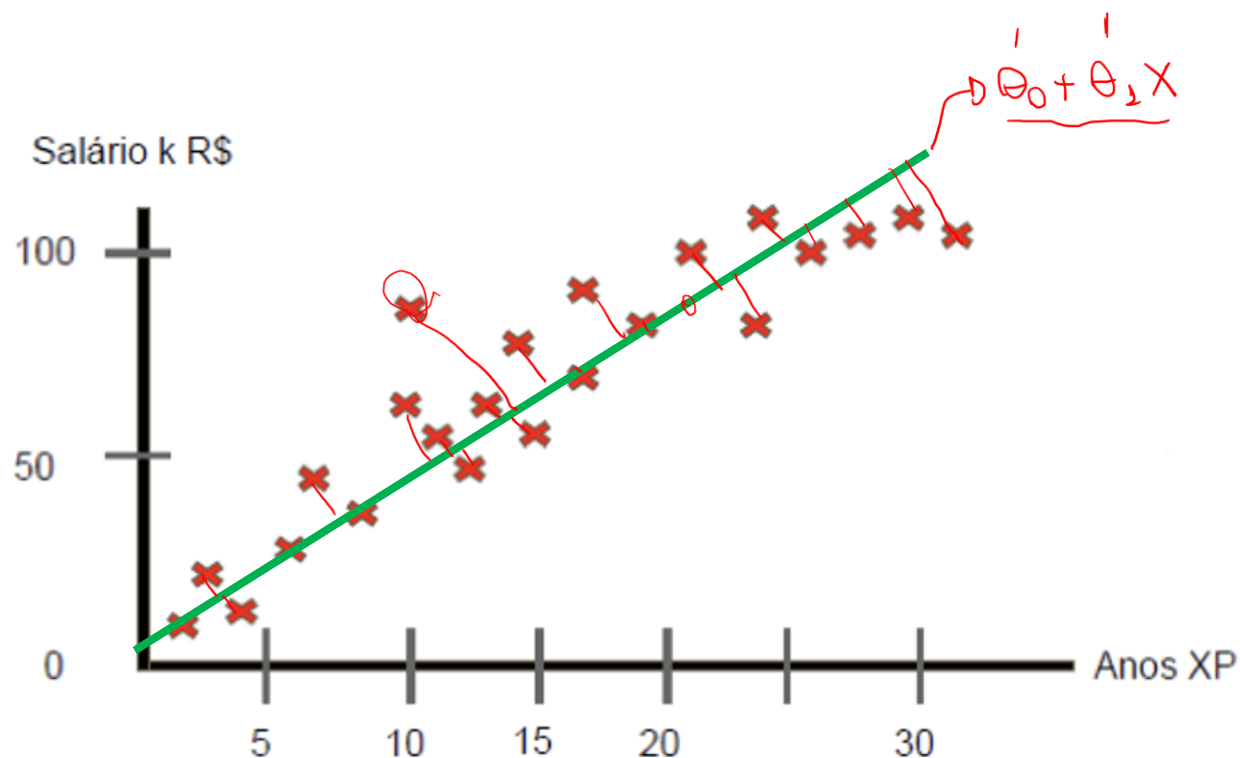
$$\theta_0 = 40 \mid \theta_1 = 0$$



$$h(x) = 10_0 + 0.5_1 \cdot x$$

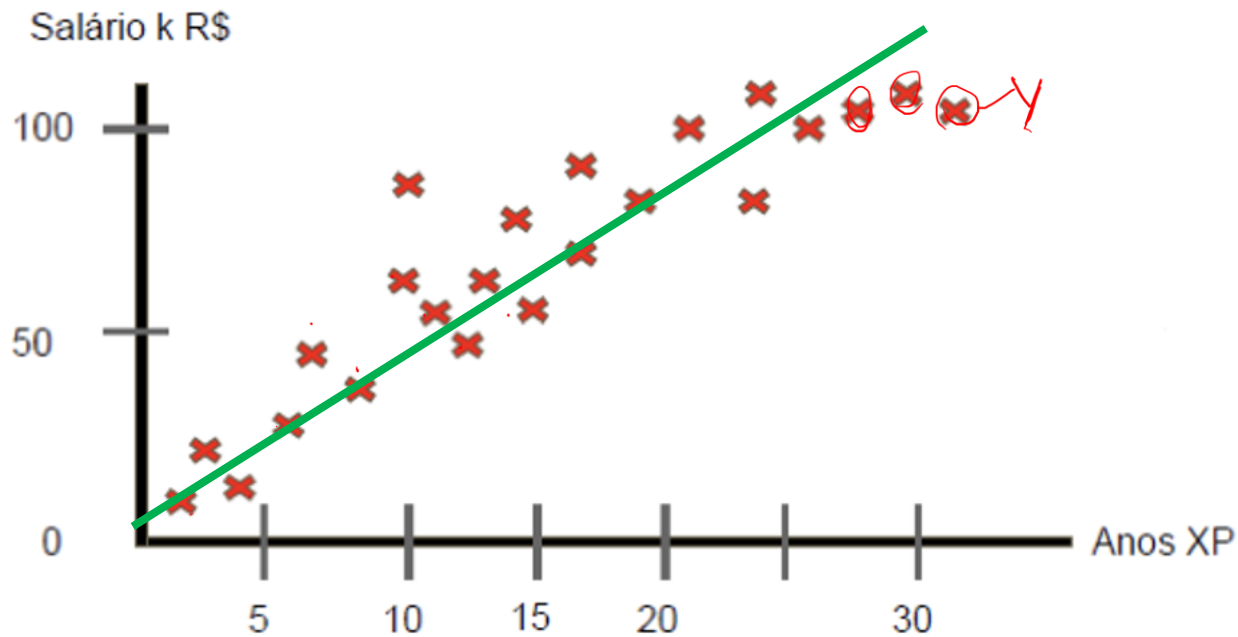
$$\theta_0 = 10 \mid \theta_1 = 0.5$$

Regressão Linear - Função de Custo



Escolher θ_0, θ_1 para que $h(x)$ seja perto de $\hat{y}$ com base no set de treinamento.

Regressão Linear - Função de Custo



Escolher θ_0, θ_1 para que $h(x)$ seja perto de y com base no set de treinamento.

Hipótese: $h(x) = \theta_0 + \theta_1 \cdot x$

Minimize: θ_0, θ_1

Mean Square Error Function

$$\sum_{i=1}^n (y^i, \hat{y}^i)^2$$

Regressão Linear – Gradiente Descendente

Objetivo: $h(x) = \theta_0 + \theta_1 \cdot x$

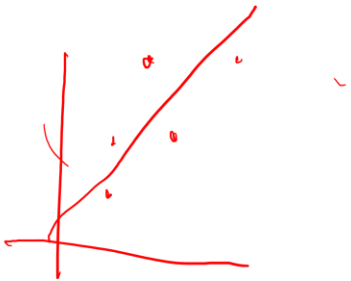
- Minimizar θ_0, θ_1

$$\theta_0 = 0 \quad \theta_1 = 0$$

$$\theta_0 = \text{random} \quad \theta_1 = \text{random}$$

Como Fazer:

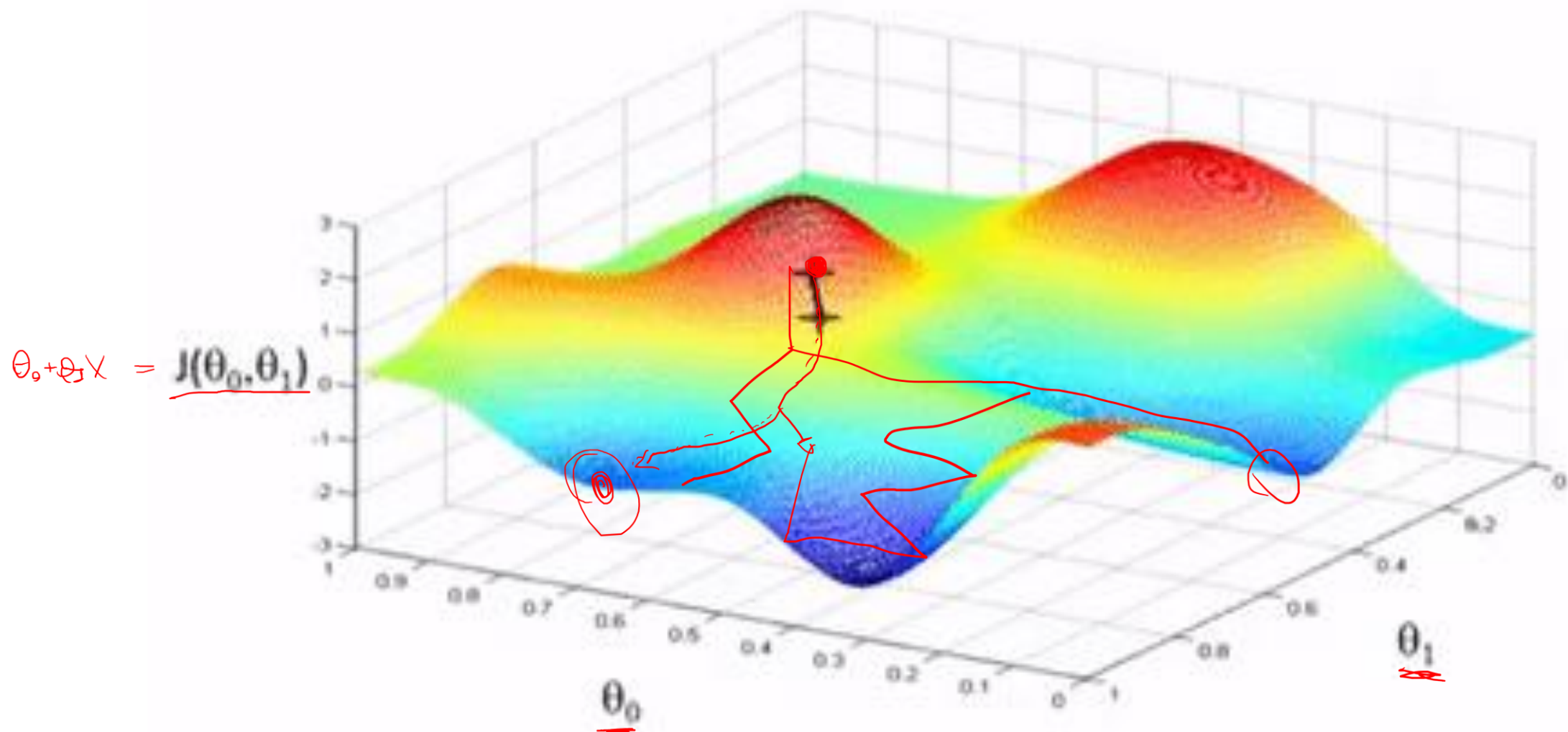
- Começar com um θ_0, θ_1 qualquer
- Mudar os valores de θ_0, θ_1 para reduzir o erro, até que se chegue a um mínimo



Mean Square Error Function

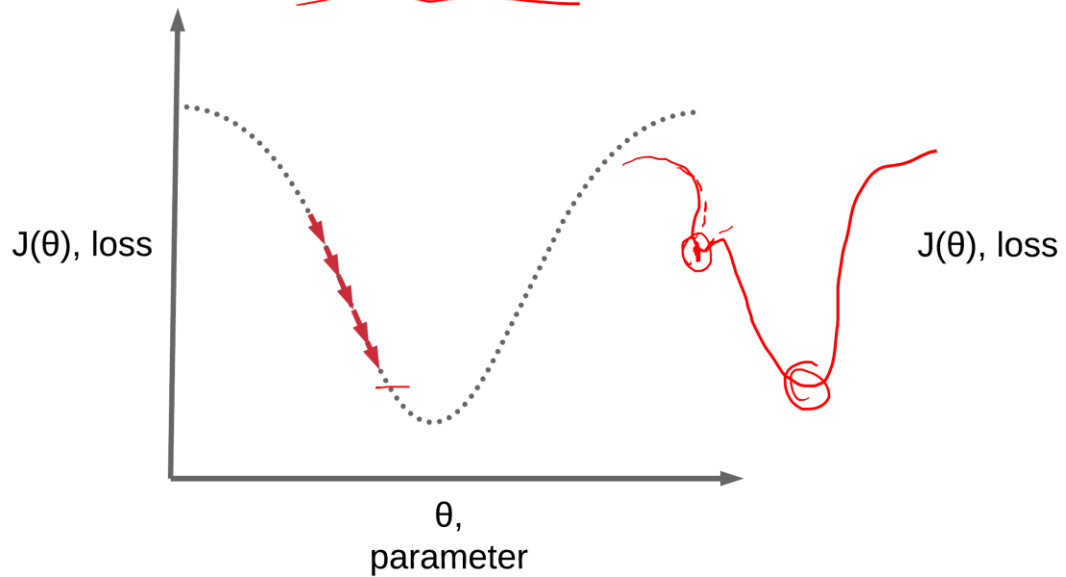
$$\sum_{i=1}^n (y^i, \hat{y}^i)^2$$

Regressão Linear – Gradiente Descendente

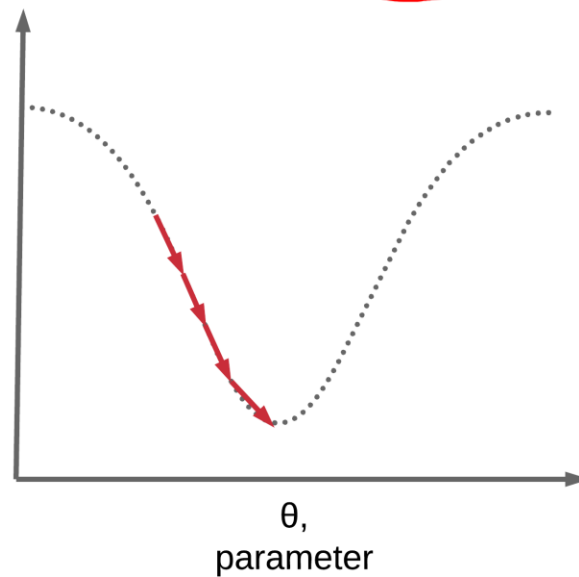


Regressão Linear – Learning Rate

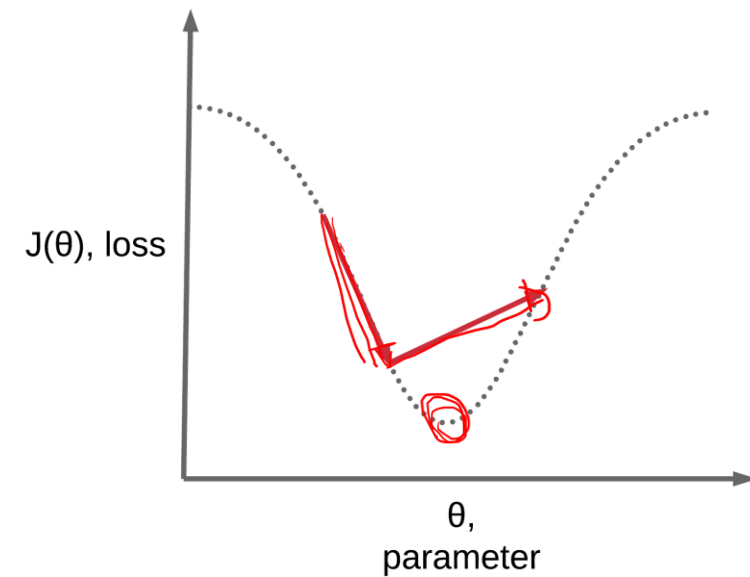
Low Learning Rate



Decent learning Rate



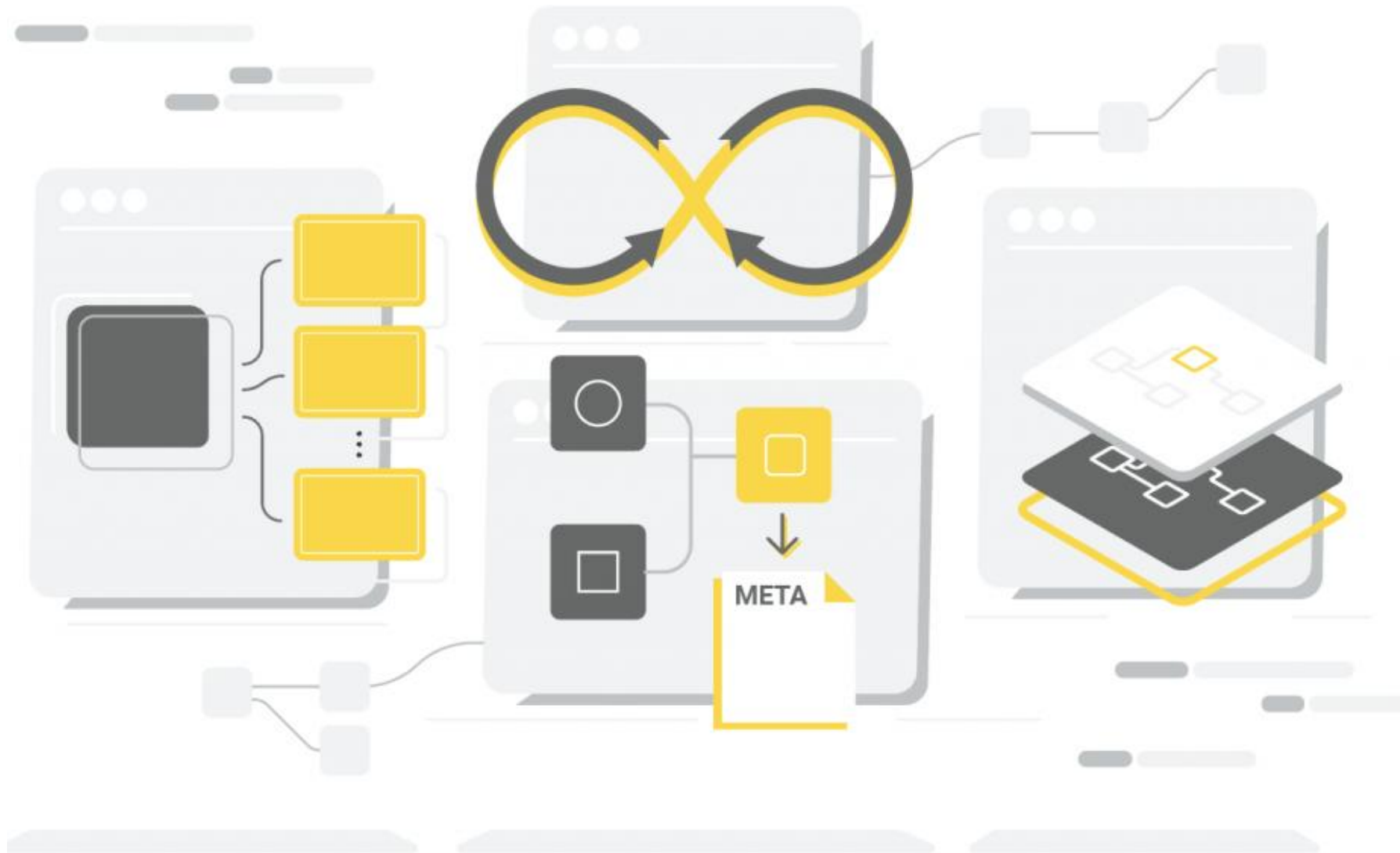
High Learning Rate





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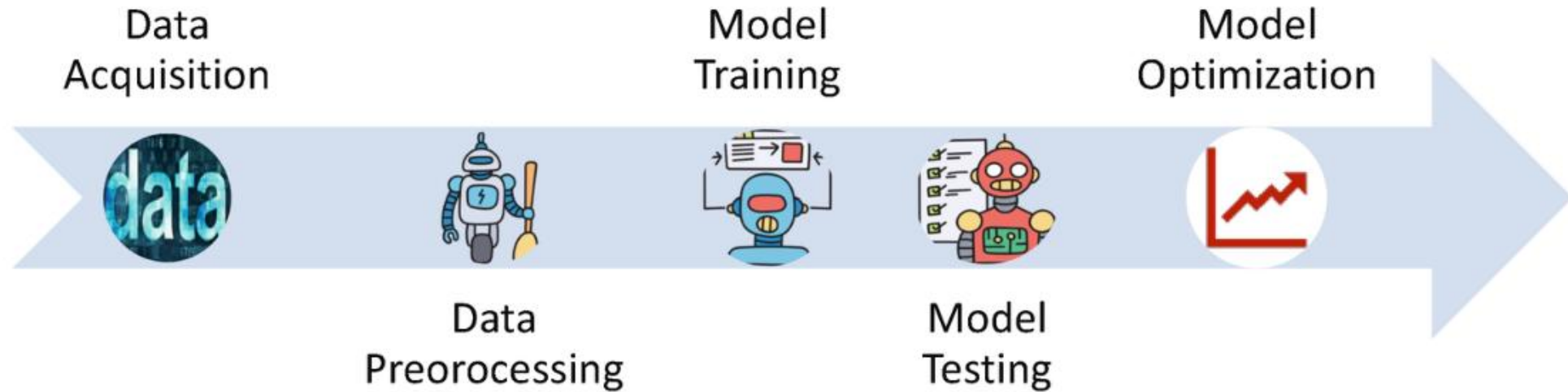




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Regressão Linear

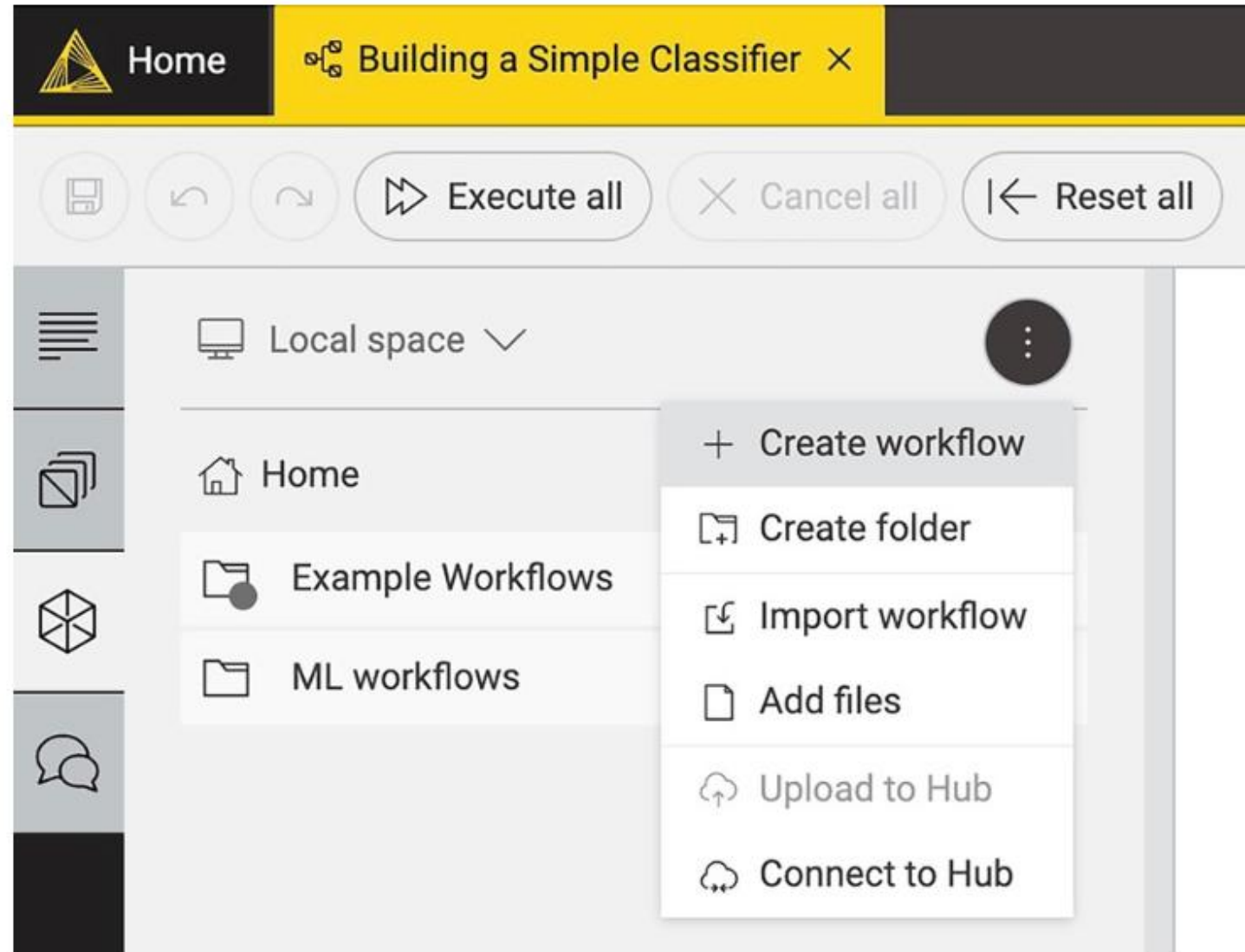




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Regressão Linear

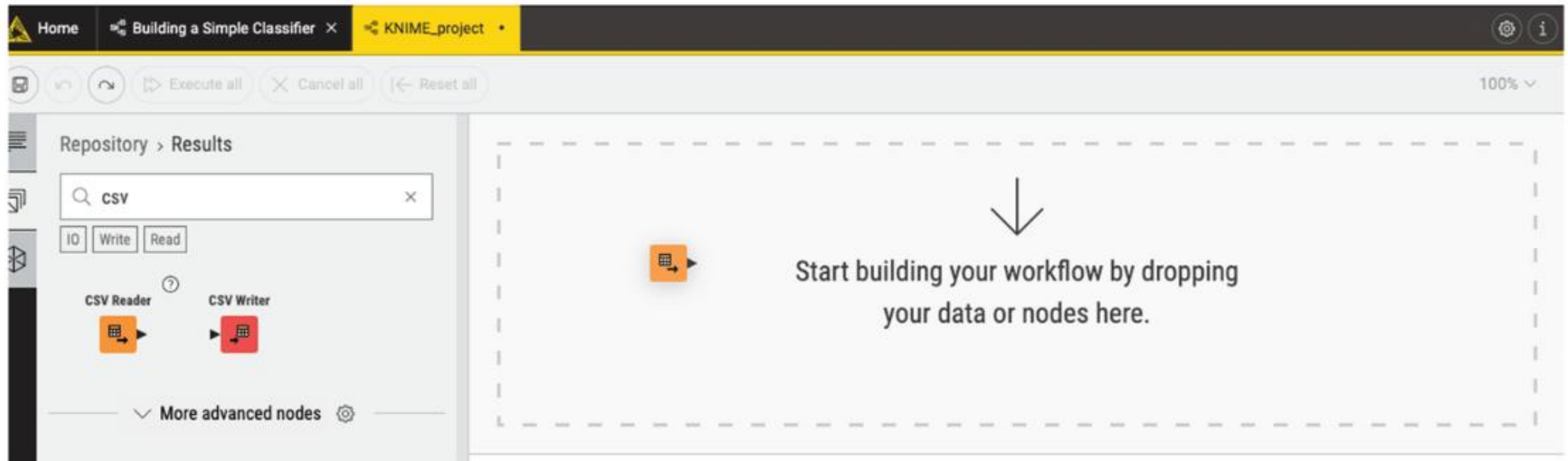




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Regressão Linear





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Regressão Linear

Dialog - 3:1 - CSV Reader

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File /Users/vcasadei/Documents/files/Salary_Data.csv Browse...

Reader options

Format

Autodetect format

Column delimiter , Row delimiter ☒ Line break ☐ Custom \n

Quote char " Quote escape char "

Comment char

☒ Has column header ☐ Has RowID

☐ Support short data rows ☐ Prepend file index to RowID

Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	D Years...	D Salary
Row0	1.1	39,343
Row1	1.3	46,205
Row2	1.5	37,731
Row3	2	43,525
Row4	2.2	39,891
Row5	2.9	56,642
Row6	3	60,150
Row7	3.2	54,445
Row8	3.2	64,445
Row9	3.7	57,189
Row10	3.9	63,218
Row11	4	55,794
Row12	4	56,957
Row13	4.1	57,081
Row14	4.5	61,111

OK Apply Cancel ?



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Regressão Linear

Home Building a Simple Classifier KNIME_project ch3.Simple Logistic Regression

Execute Cancel Reset Create metanode Create component 119%

Repository

Search Nodes

Table Cropper Cell Extractor Show all

Views

Bar Chart Line Plot Pie Chart

Stacked Area Chart **Scatter Plot** Statistics

Heatmap Histogram Show all

Other Data Types

CSV Reader

Scatter Plot

Add comment

Flow Variables

Count: 1

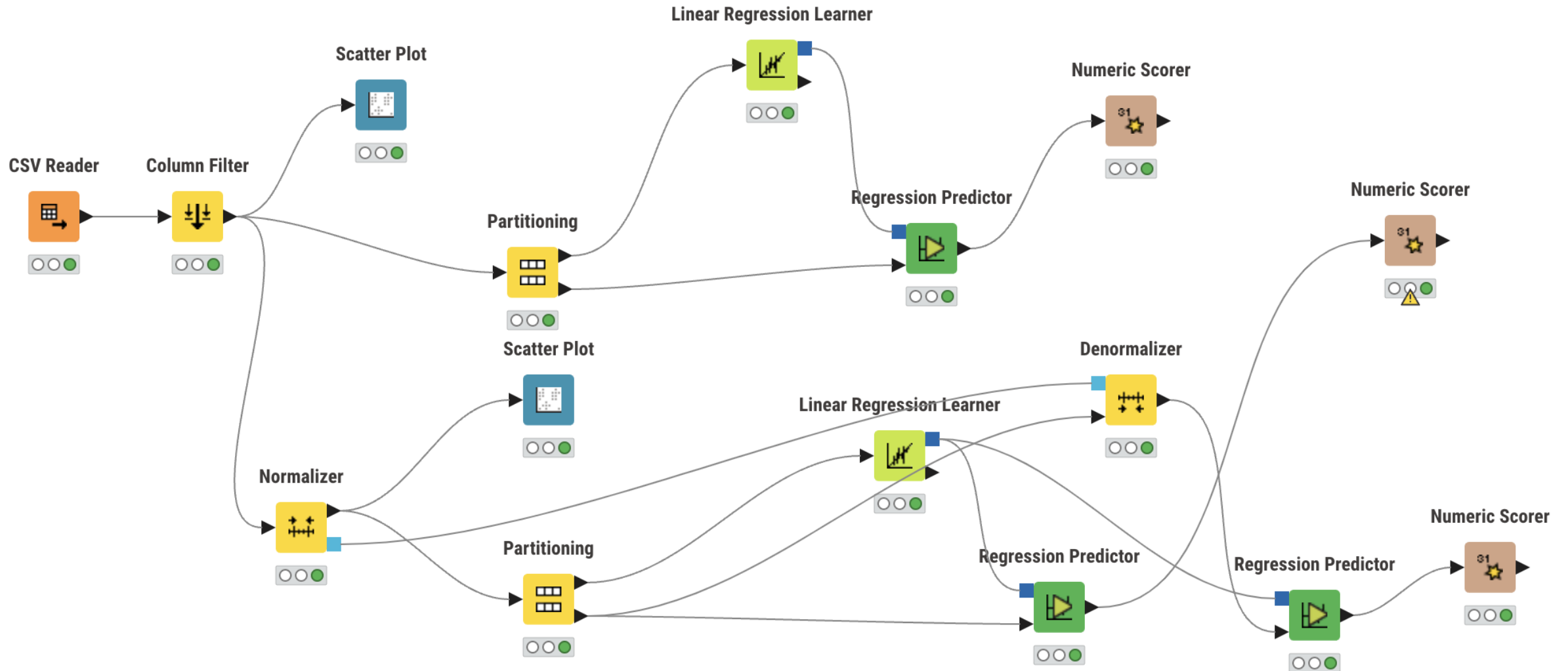
Owner ID	Data Type	Variable Name	Value
	StringType	knime.workspace	/Users/Mrk/knime5_workflows



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Regressão Linear



Regressão Logística

Classificação:

- y é um valor discreto

Classificação
Categoria

V/F

Gato / Cachorro

Câncer / Não-Câncer

A - B - C - D - Z - 0 - 1 9

37

Exemplos:

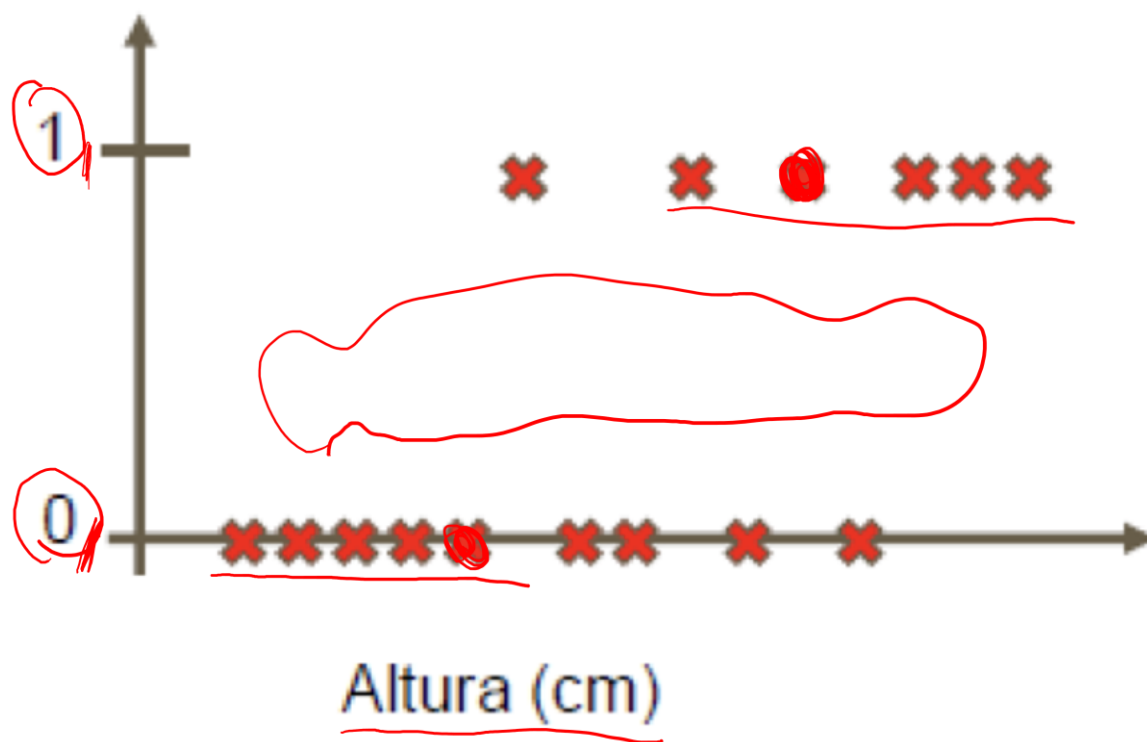
- Email: Spam/Não Spam
- Fraudes em Transações
- Detecção de tumor

Y:

- 0
- 1

Regressão Logística

Pessoa Adulto

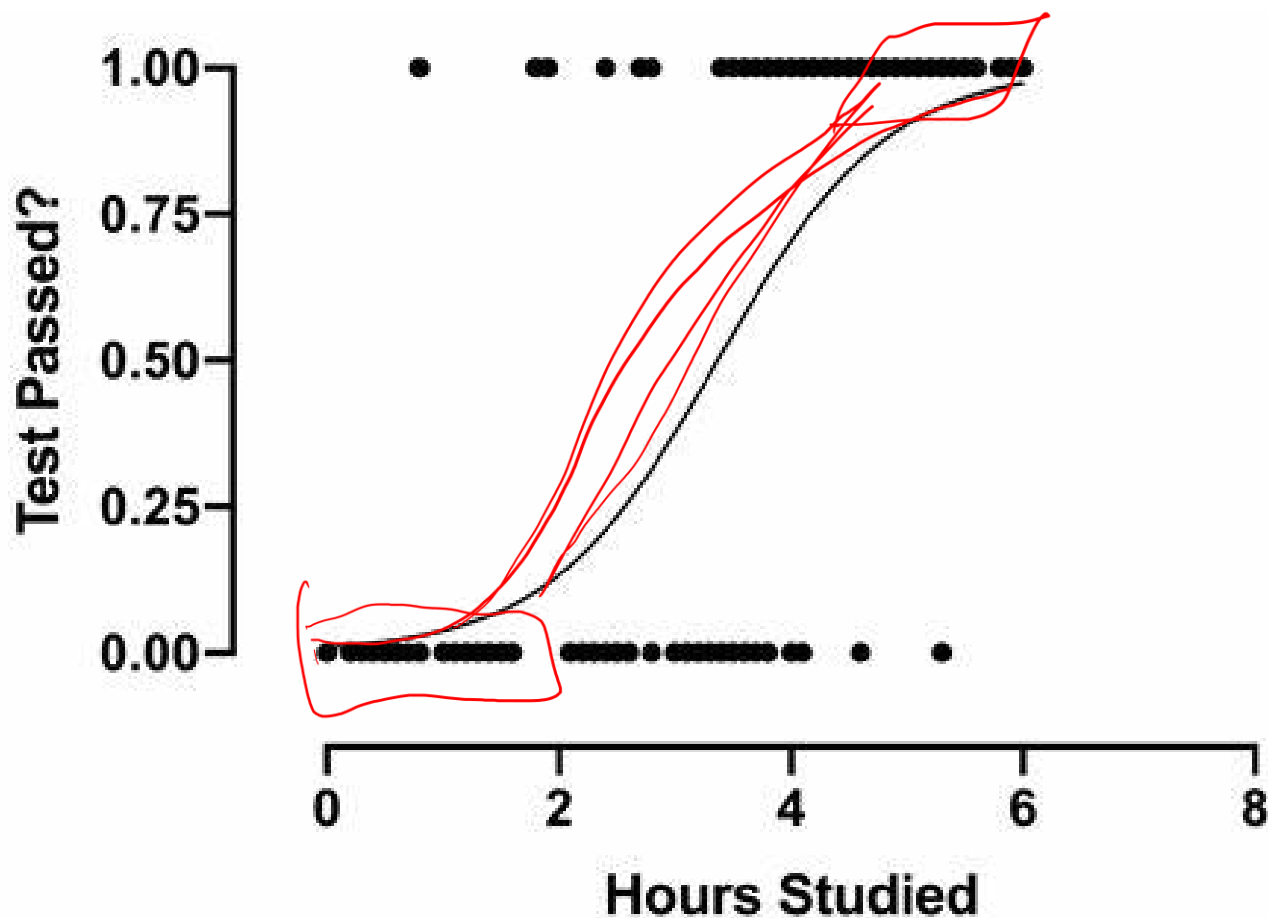


Classificação: Predição de Valores Discretos

Não é uma boa ideia usar Regressão Linear para esse tipo de problema

Regressão Logística

Linear $\Rightarrow h(x) = \theta_0 + \theta_1 x$



$$\Theta^T = [\theta_0 \ \theta_1]$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} \begin{bmatrix} x_0 & x_1 & \dots & x_n \end{bmatrix}$$

Objetivo:

- $0 \leq h(x) \leq 1$
- $h(x) = g(\theta^T \cdot X)$
- Sigmoid Function

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T \cdot X}}$$

Regressão Logística – Função de Custo

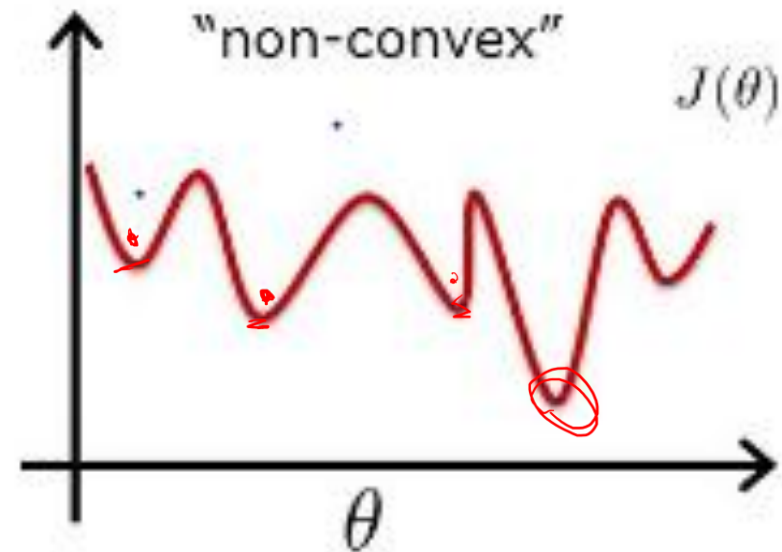
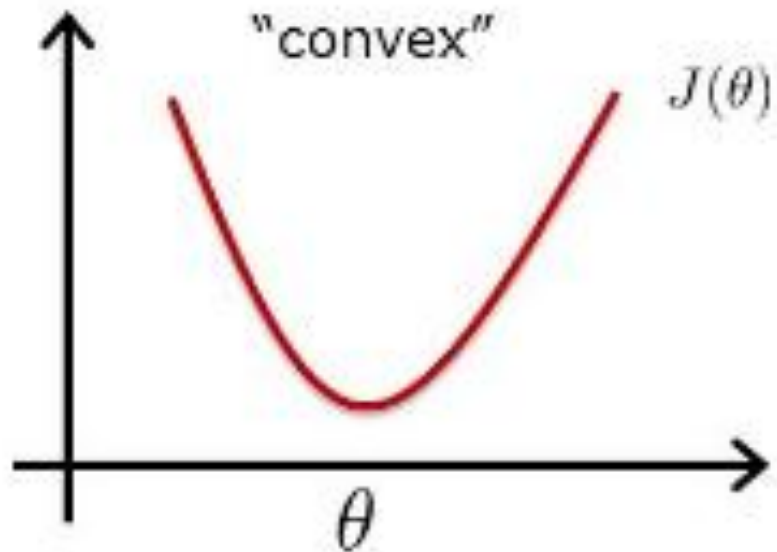


Não podemos usar MSE

"... because our prediction function is non linear (due to sigmoid transform). Squaring this prediction as we do in MSE results in a non convex function with many local minimums. If our cost function has many local minimums, gradient descent may not find the optimal global minimum."

Regressão Logística – Função de Custo

Convex Vs Non-Convex



Regressão Logística – Função de Custo

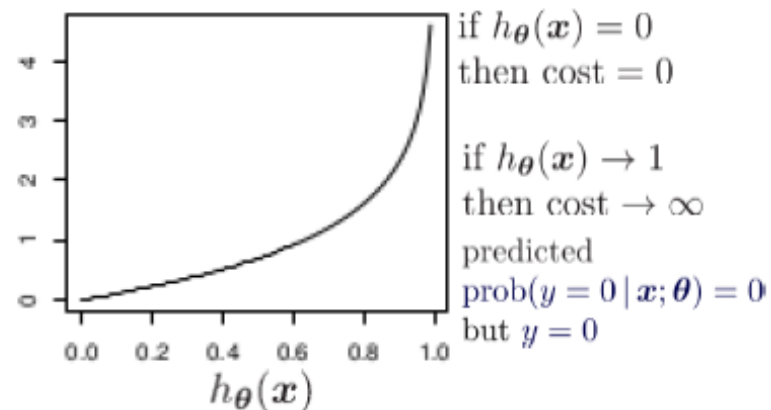
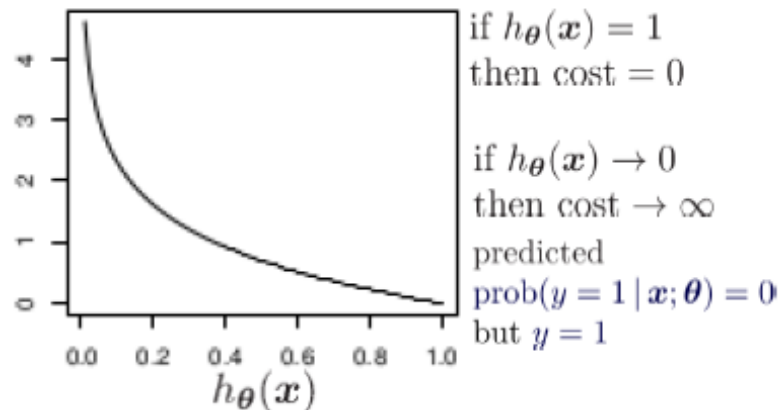
Cost Function for Logistic Regression

$$\text{Linear} \Rightarrow \sum (y_i - \hat{y}_i)^2$$

"The key thing to note is the cost function penalizes confident and wrong predictions more than it rewards confident and right predictions! The corollary is increasing prediction accuracy (closer to 0 or 1) has diminishing returns on reducing cost due to the logistic nature of our cost function."

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)), & y = 1 \\ -\log(1 - h_{\theta}(x)), & y = 0 \end{cases}$$

if $y = 1$ if $y = 0$

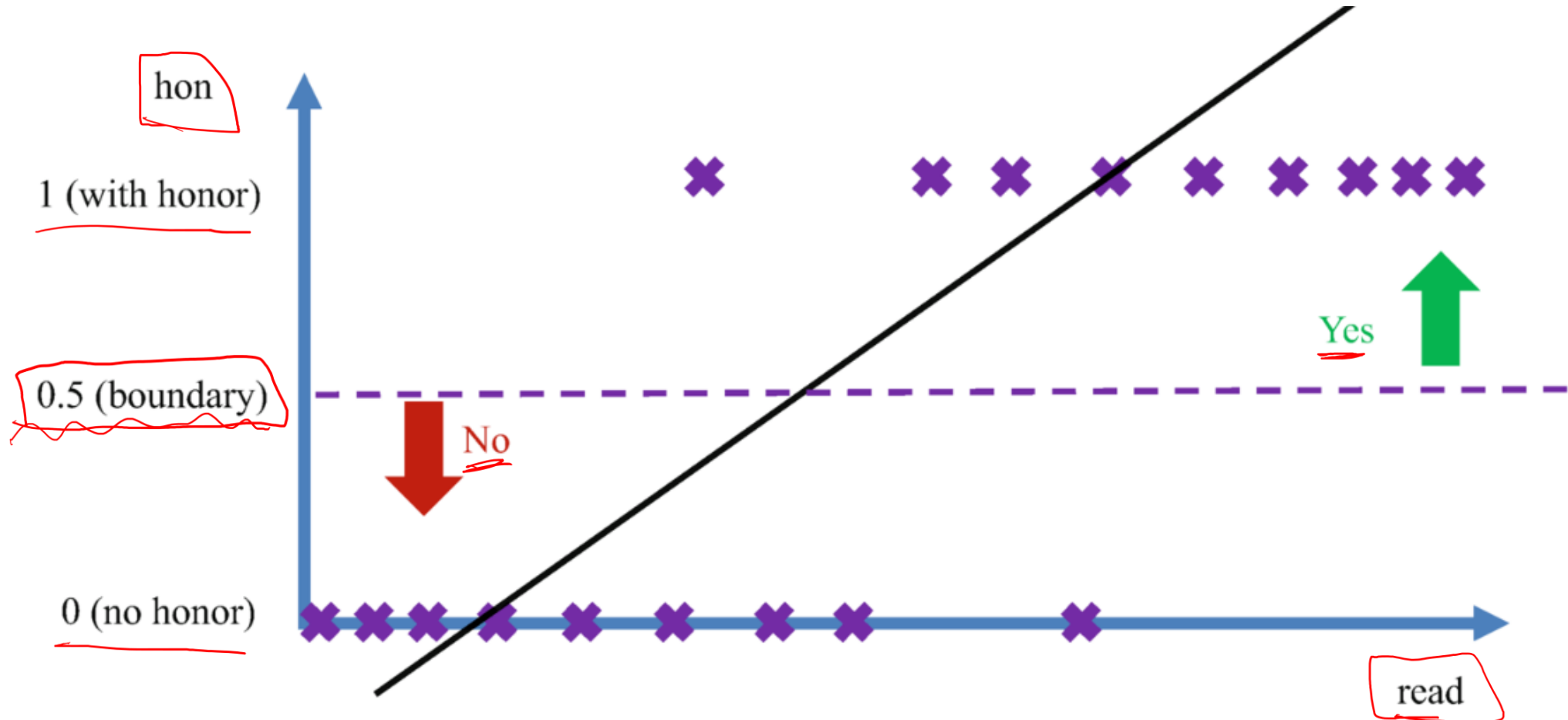




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Regressão Logística

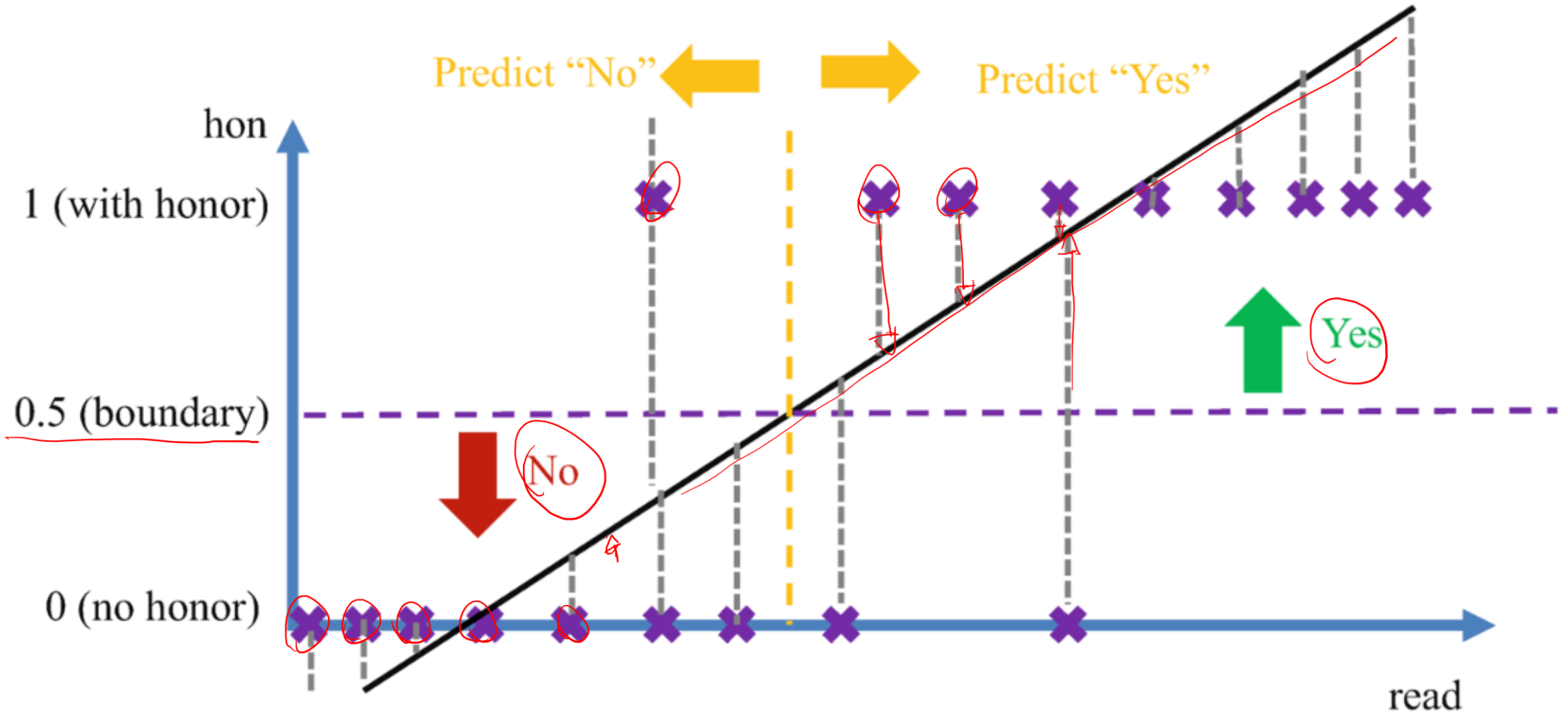




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Regressão Logística

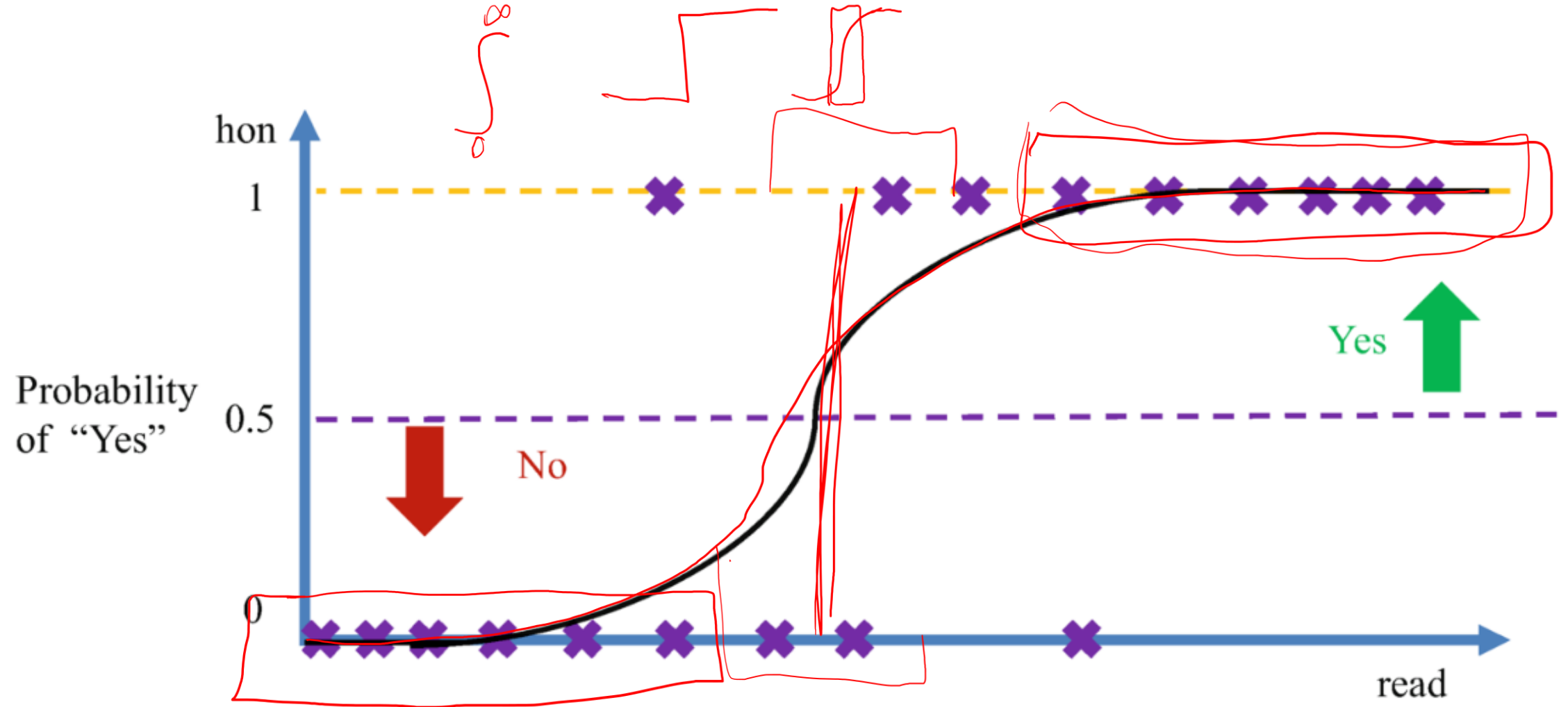




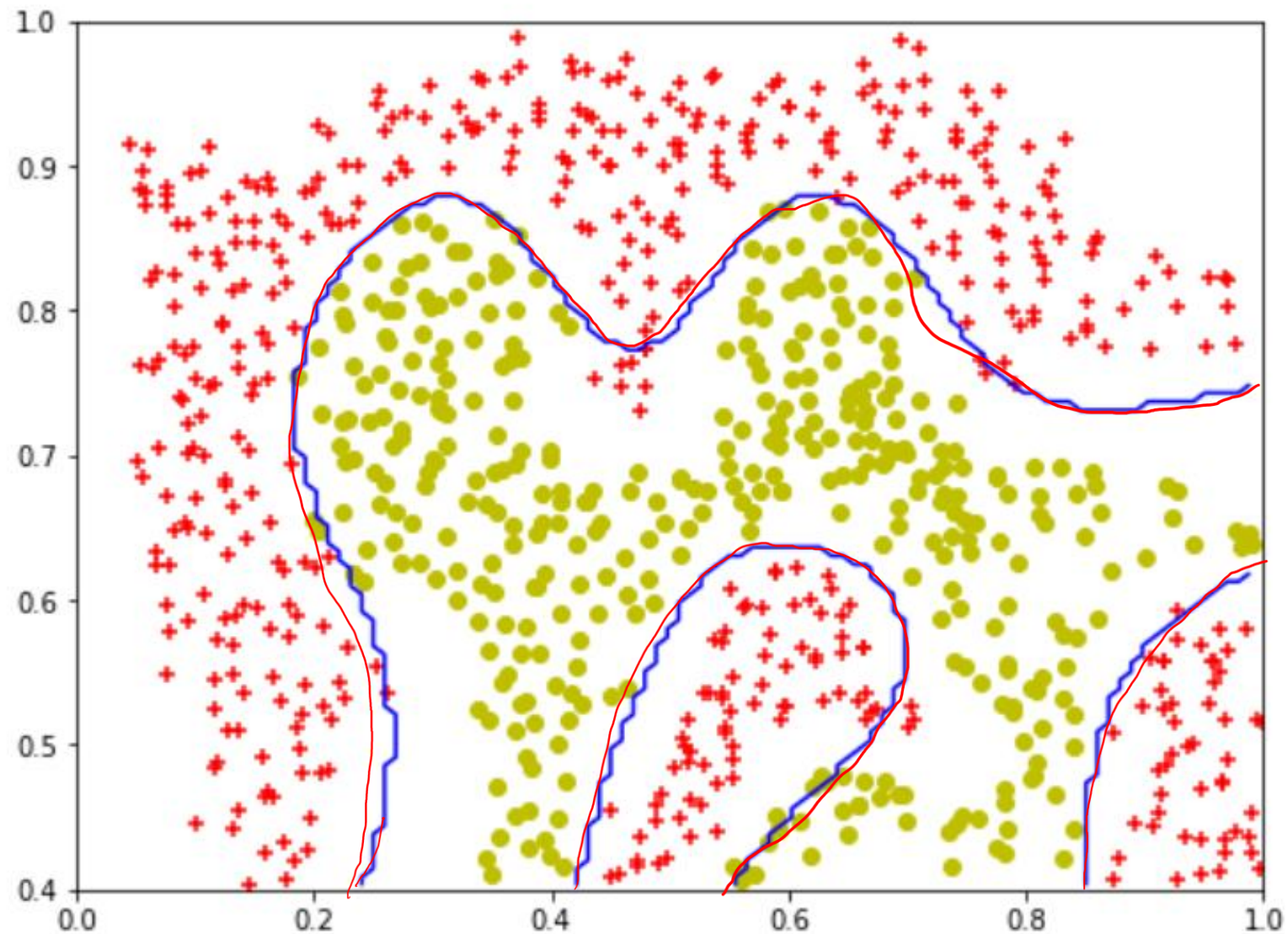
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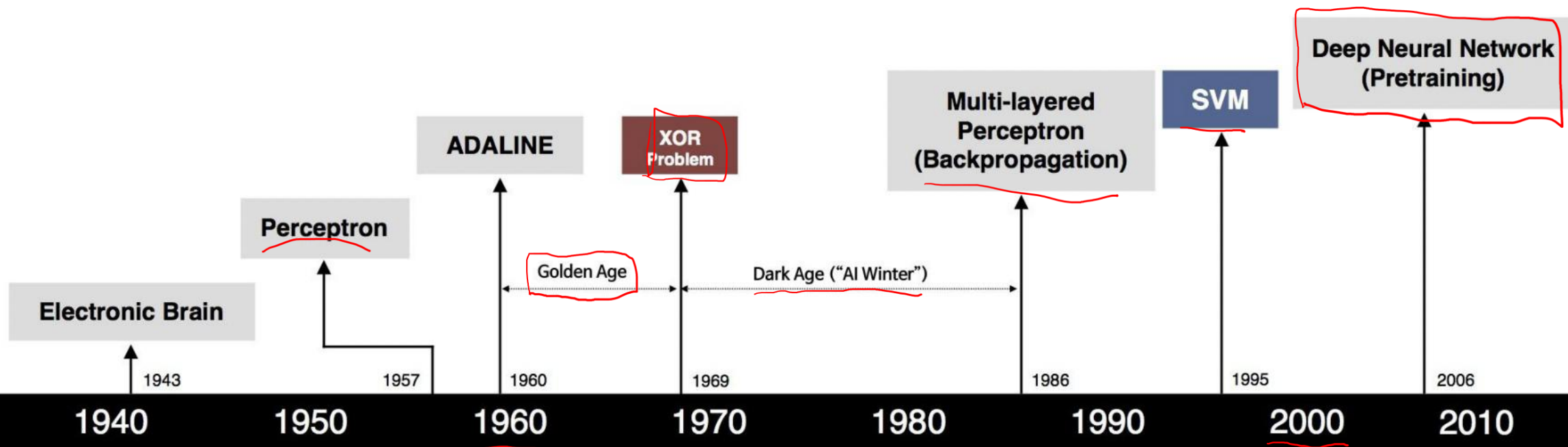
Regressão Logística



Redes Neurais - Hipótese não-linear



Redes Neurais - História



S. McCulloch - W. Pitts



F. Rosenblatt



B. Widrow - M. Hoff



M. Minsky - S. Papert



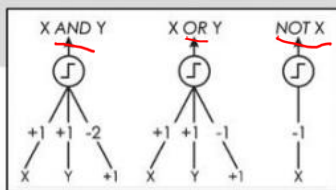
D. Rumelhart - G. Hinton - R. Williams



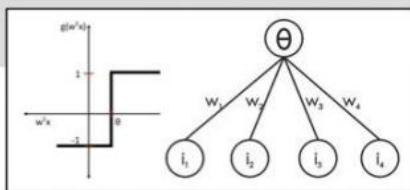
V. Vapnik - C. Cortes



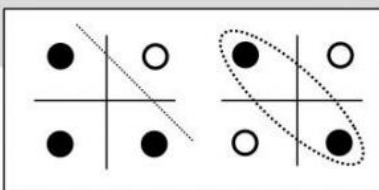
G. Hinton - S. Ruslan



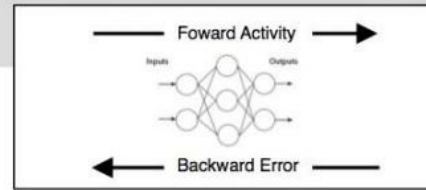
- Adjustable Weights
- Weights are not Learned



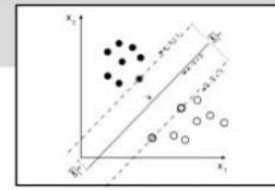
- Learnable Weights and Threshold



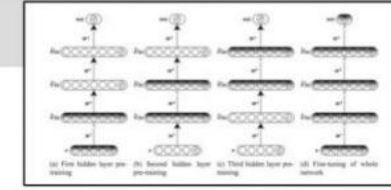
- XOR Problem



- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting

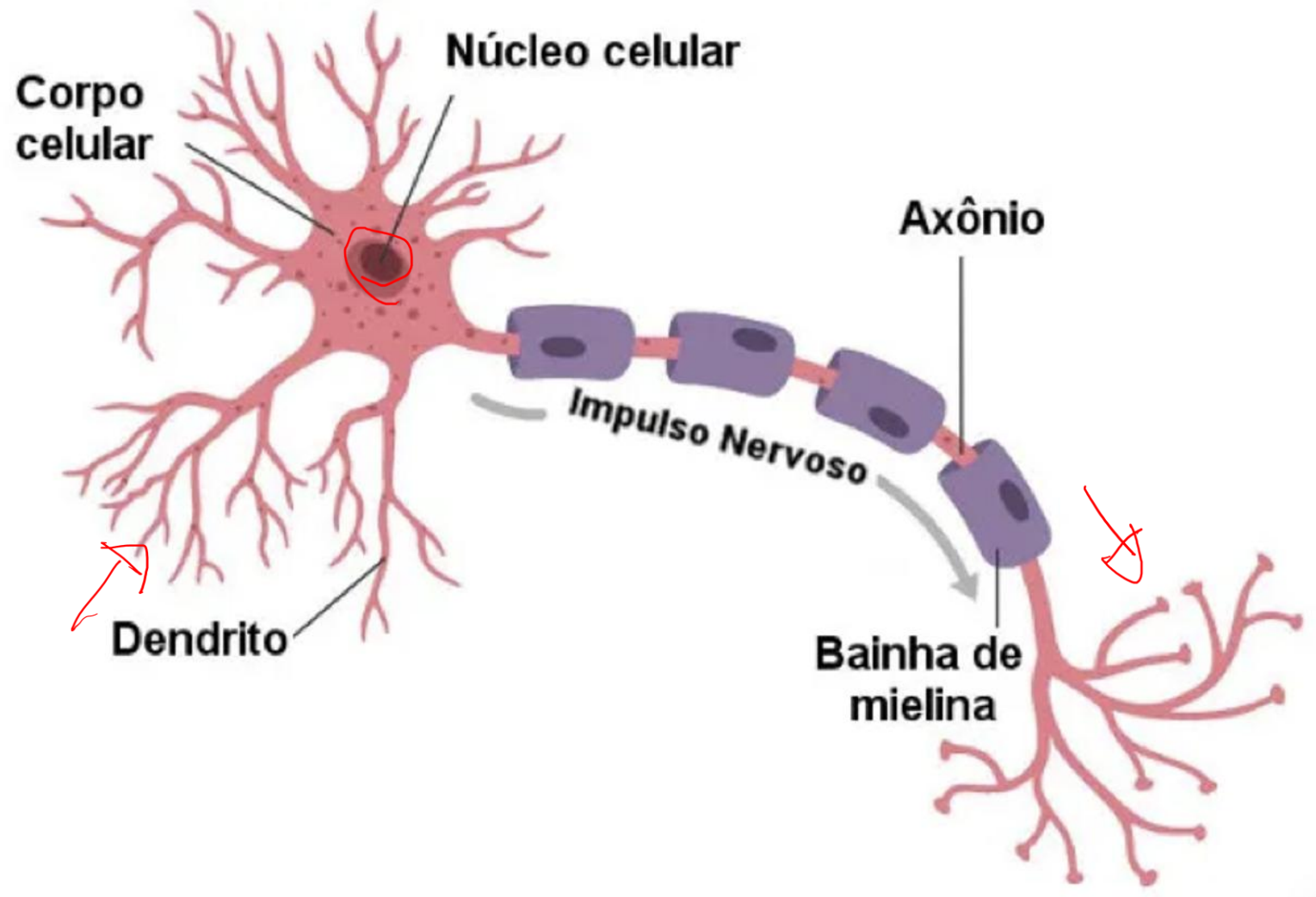


- Limitations of learning prior knowledge
- Kernel function: Human Intervention

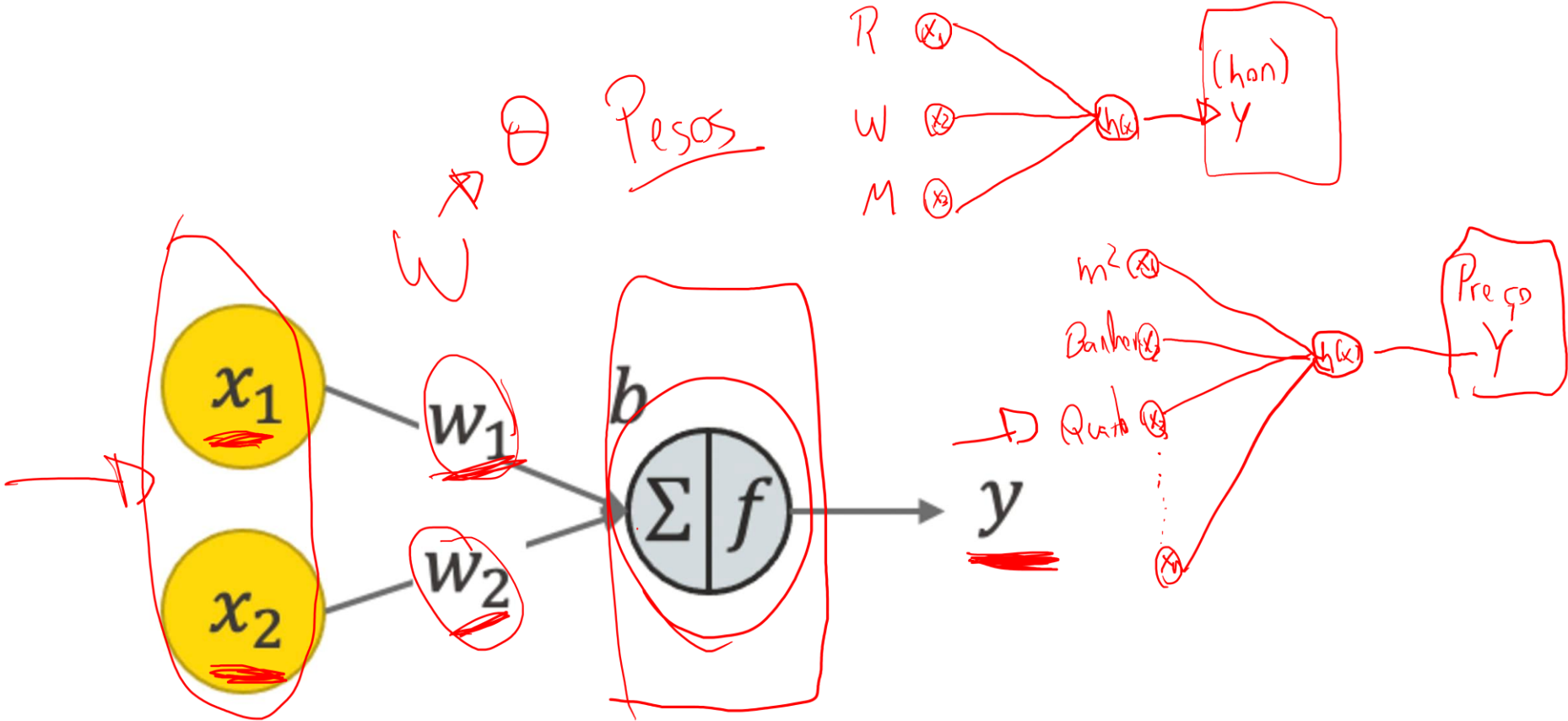


- Hierarchical feature Learning

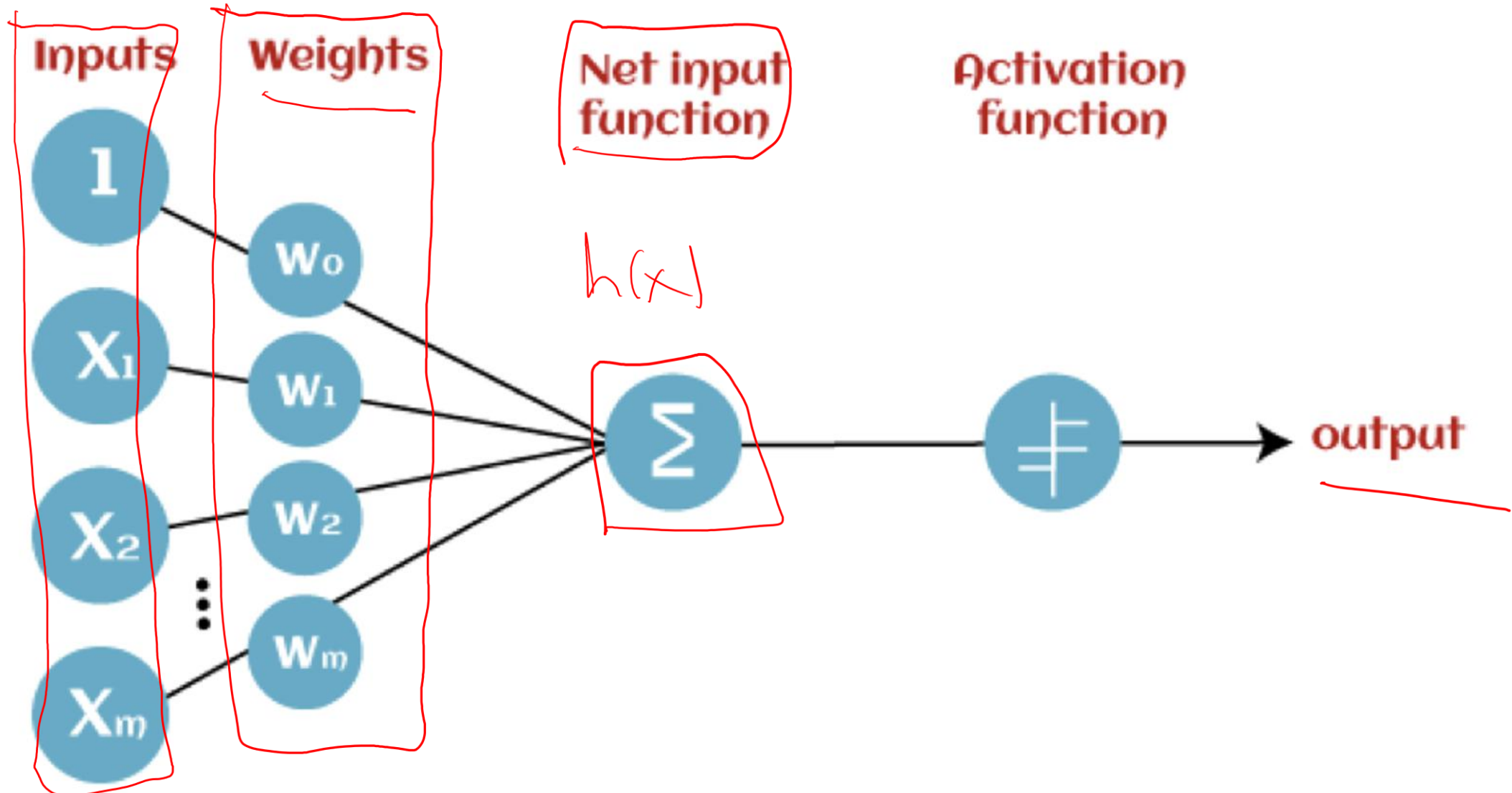
Redes Neurais - Neurônio



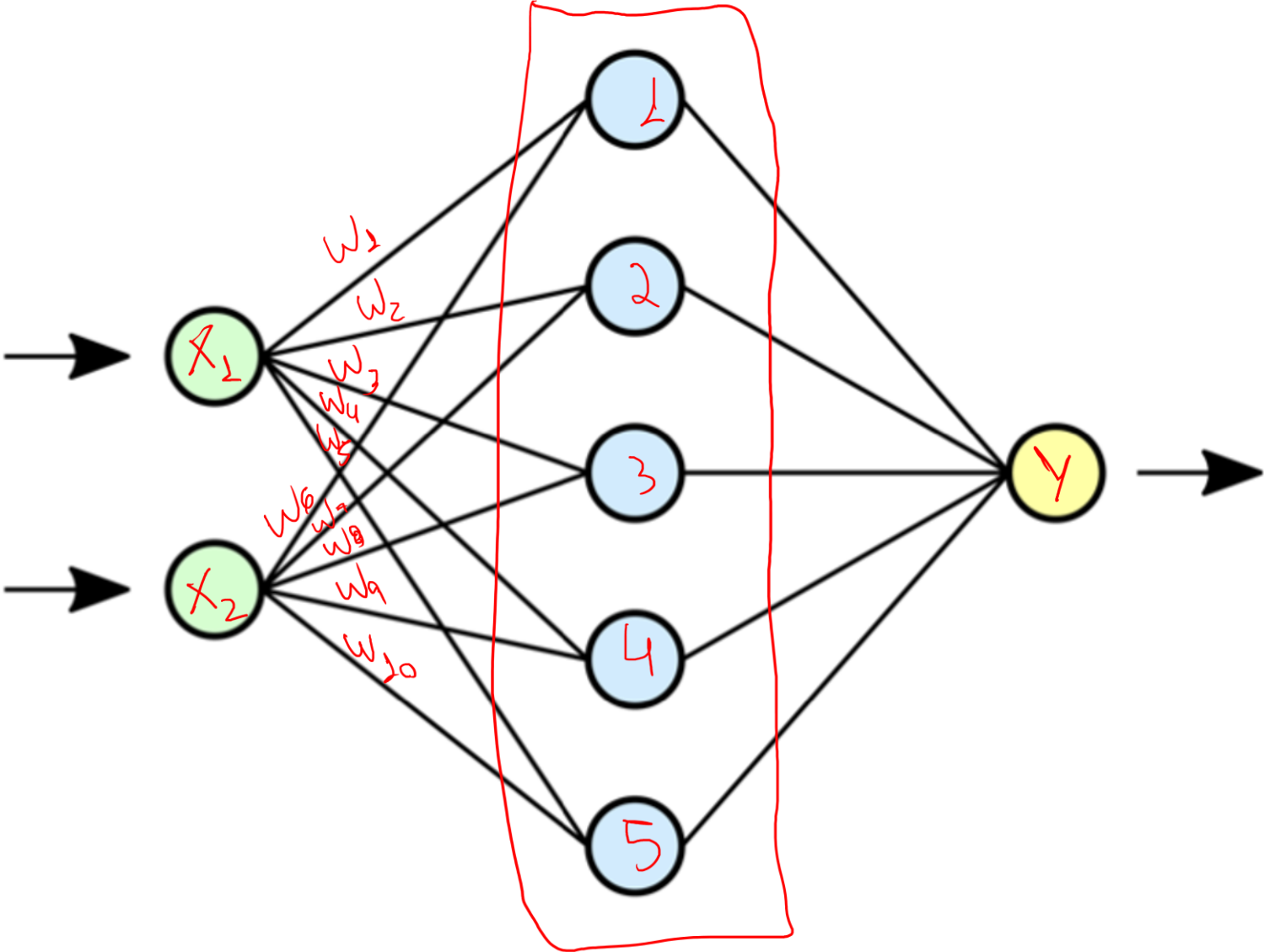
Redes Neurais - Perceptron



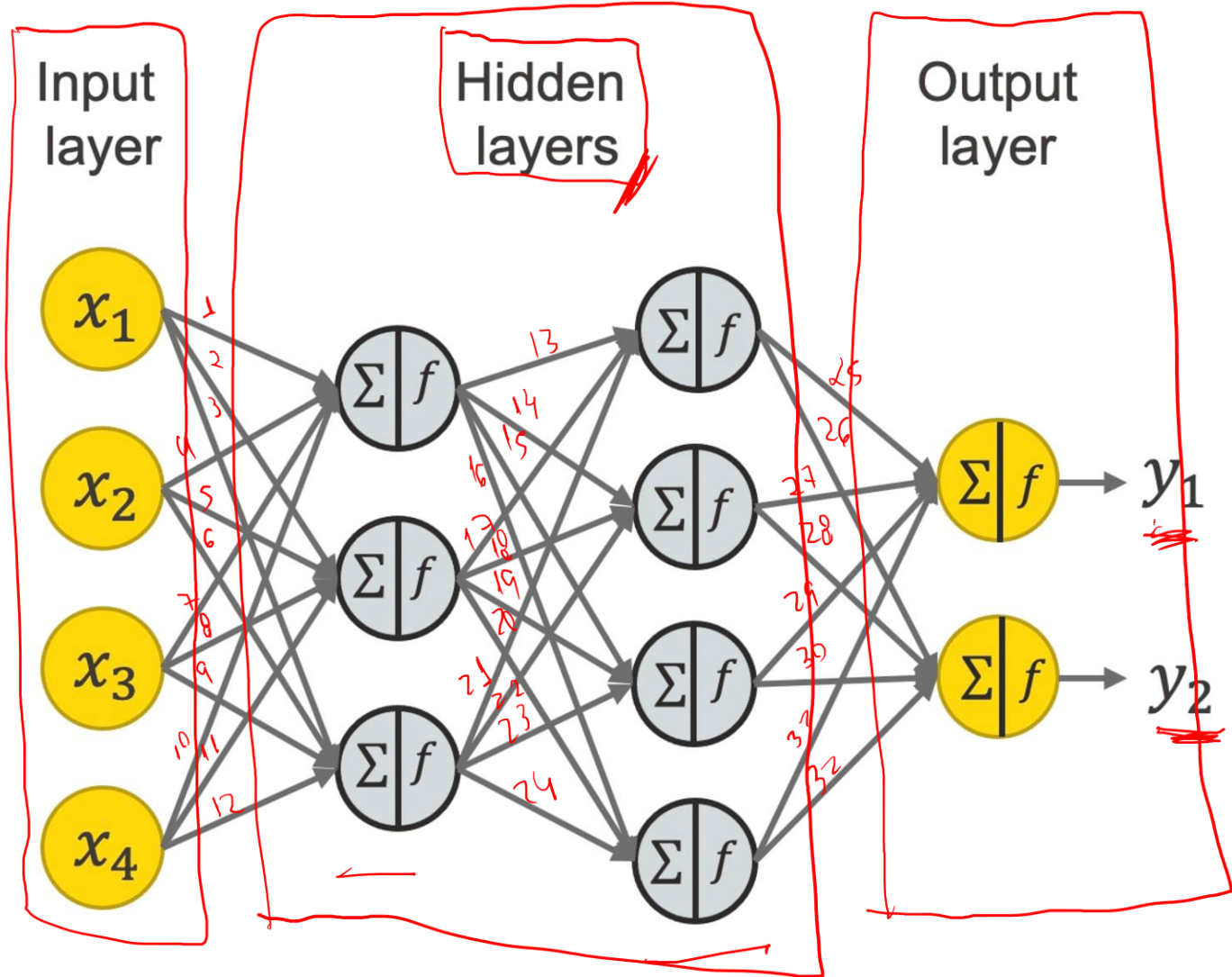
Redes Neurais – Perceptron



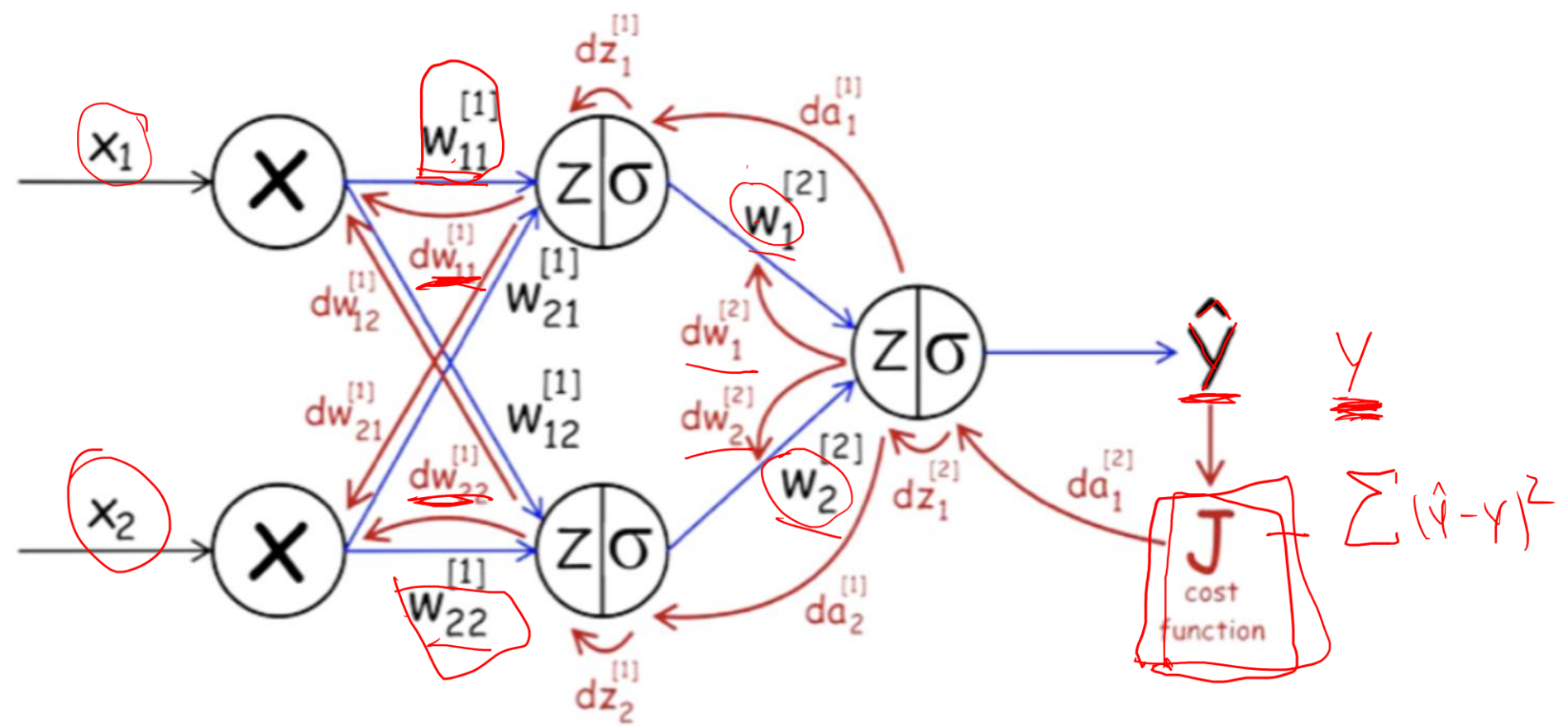
Redes Neurais



Redes Neurais

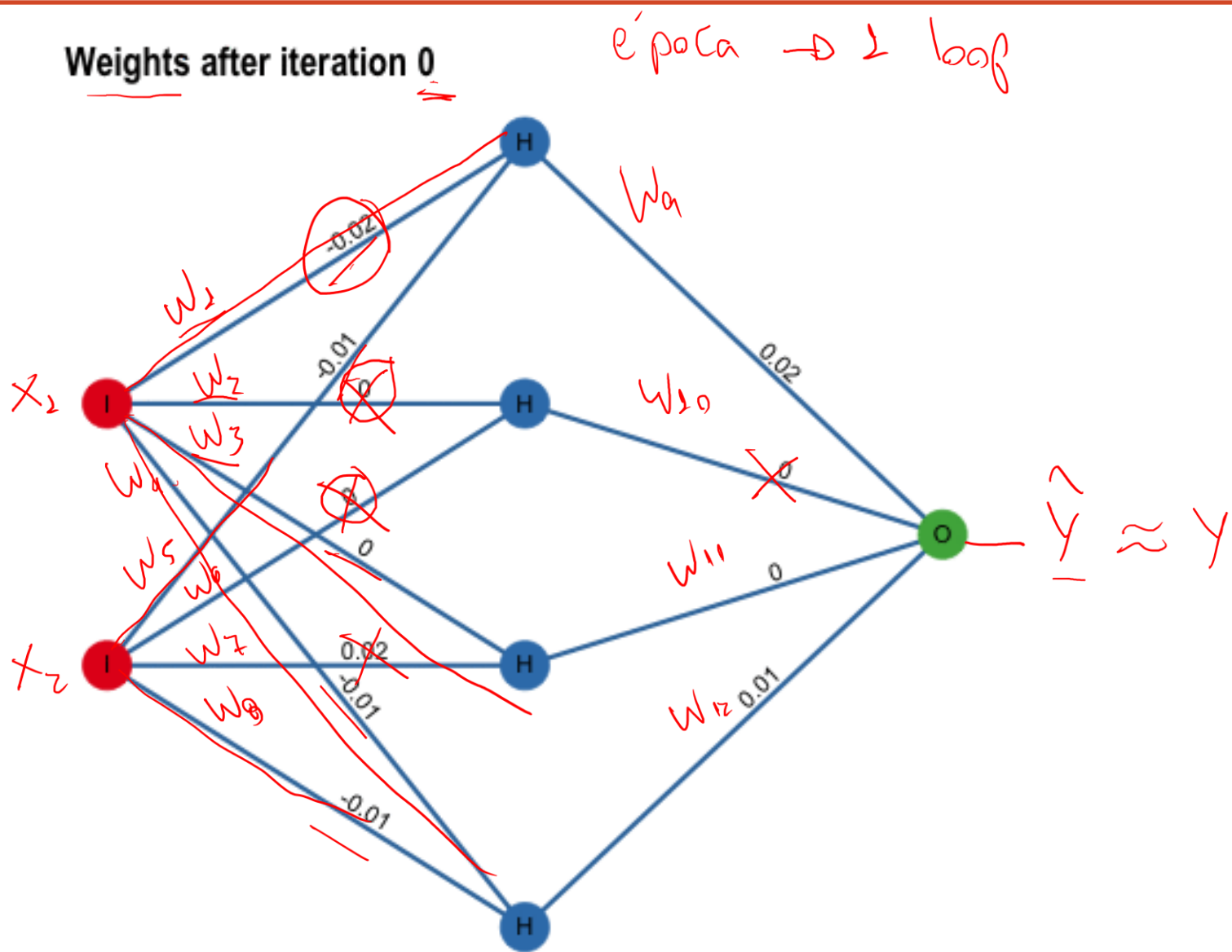


Redes Neurais - Backpropagation

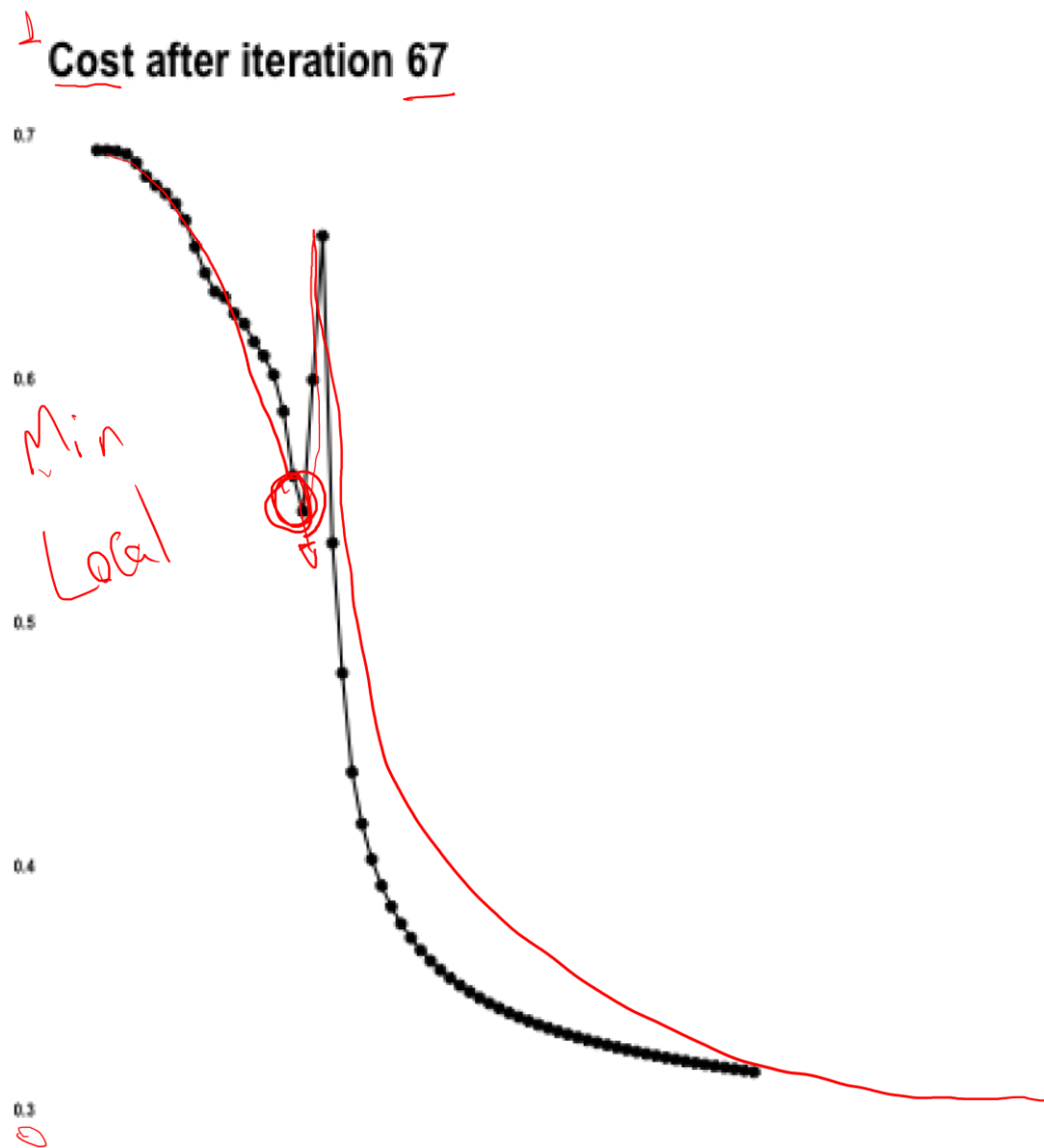


Backpropagation

Redes Neurais - Training

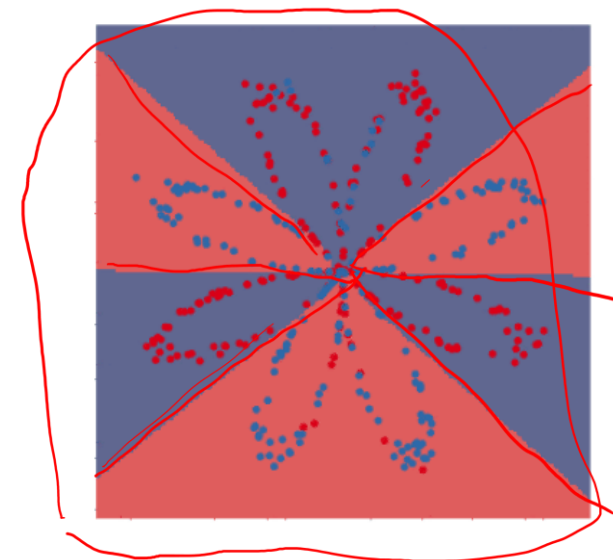
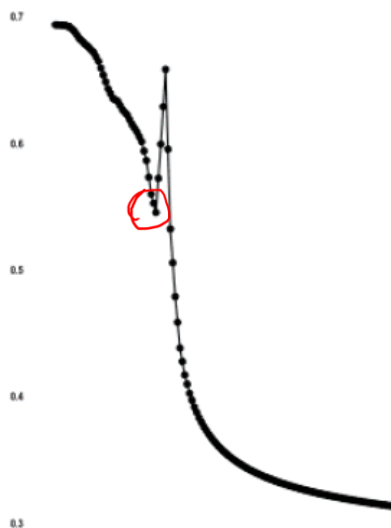
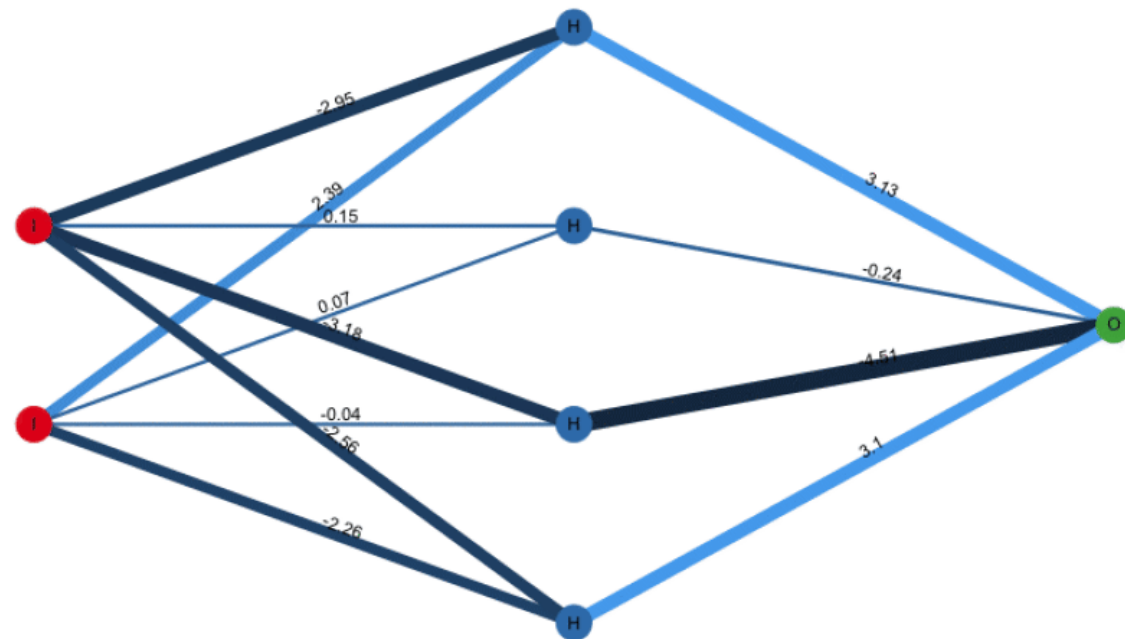


Redes Neurais - Training



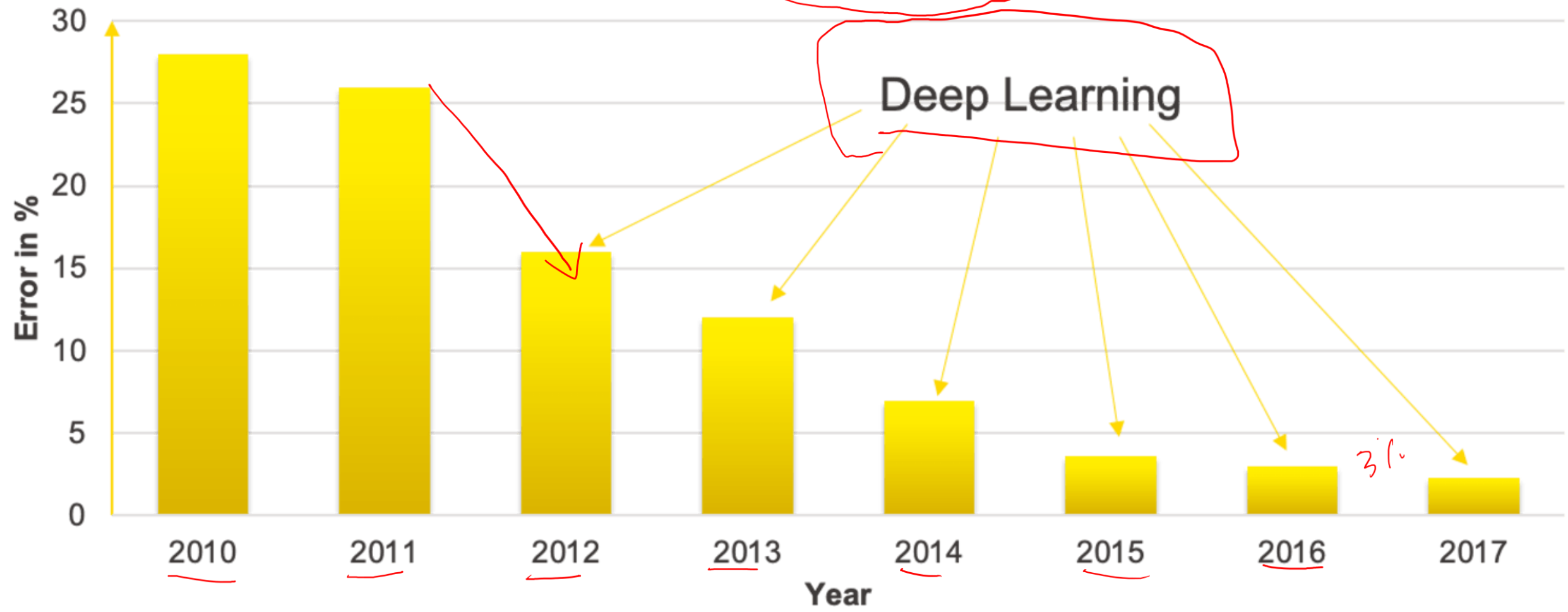
Redes Neurais - Training

Training a neural net at iteration 71

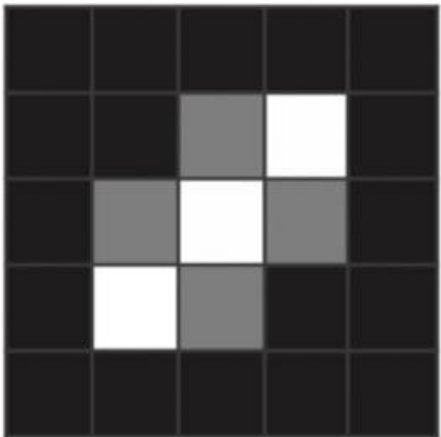


Deep Learning

Winners of the ImageNet Challenge



Deep Learning



=

-1	-1	-1	-1	-1
-1	-1	0.2	1	-1
-1	0.2	1	0.2	-1
-1	1	0.2	-1	-1
-1	-1	-1	-1	-1

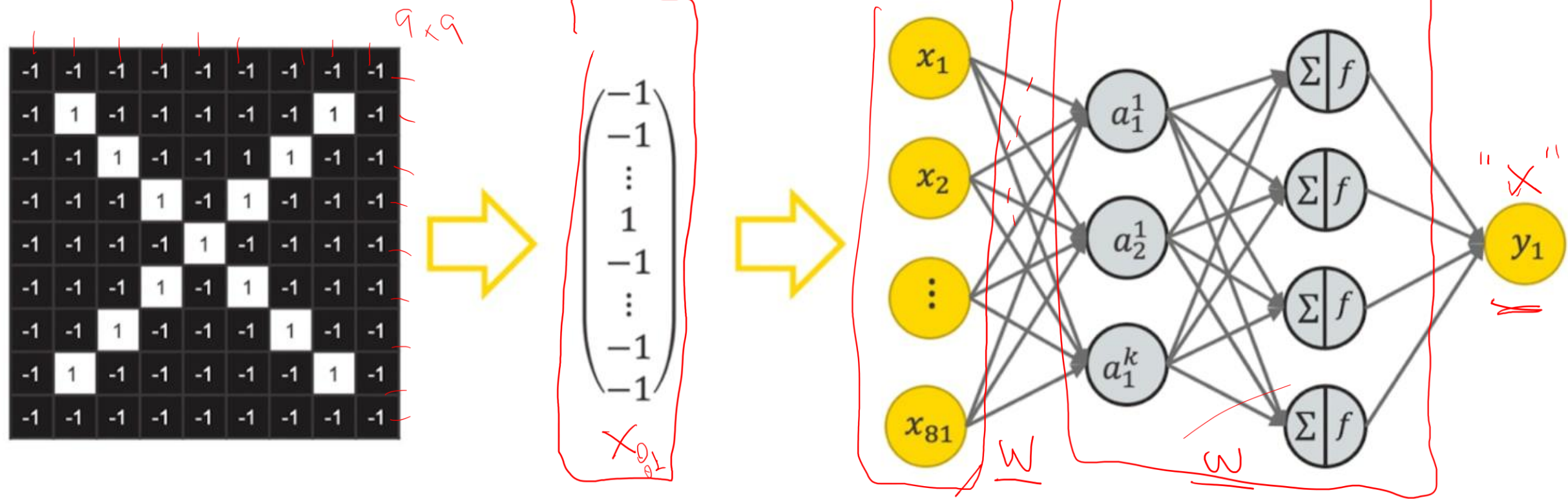
$X = \begin{bmatrix} R & W & M \\ x & x & x \\ x & x & x \\ x & x & x \\ \vdots \\ x & x & x \end{bmatrix}$



=

0.8	0.6	...	0.6	0.9		
0.1	0.5	...	0.5	0.2		
⋮	0.1	0.1	...	0.1	0.3	
0.8	0.2	0.3	0.2	...	0.1	0.1
0.6	⋮	0.5	⋮	...	⋮	⋮
0.1	0.8	0.7	...	0.8	0.8	
0.2	0.8	0.6	...	0.6	0.9	

Deep Learning



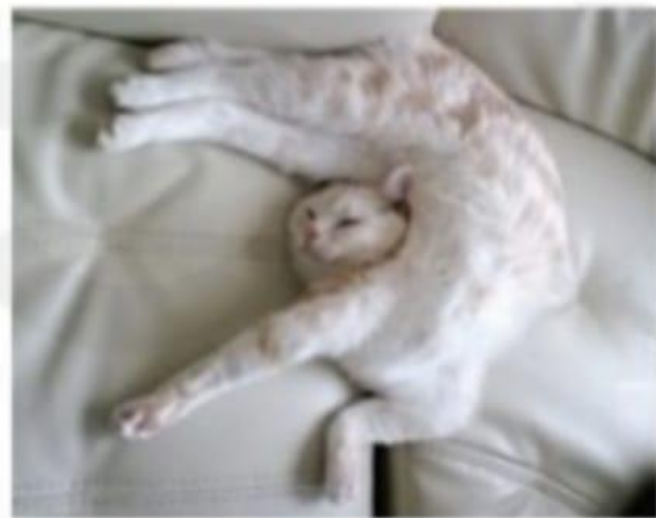
Viewpoint variations



Illumination conditions



Deformations

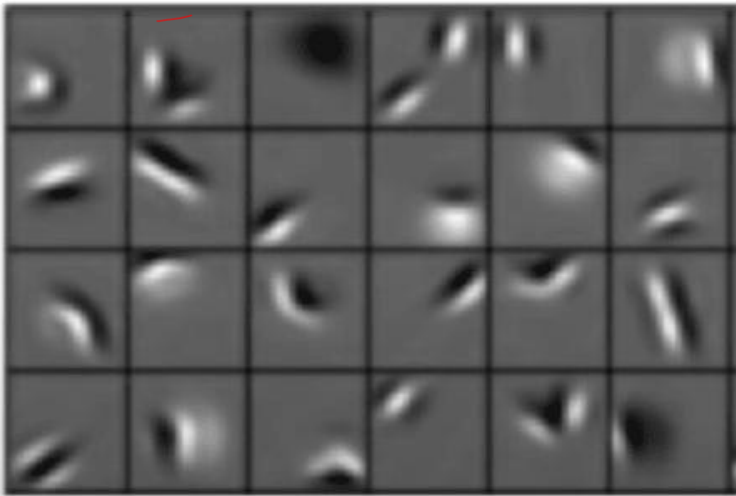


Intra-class variations



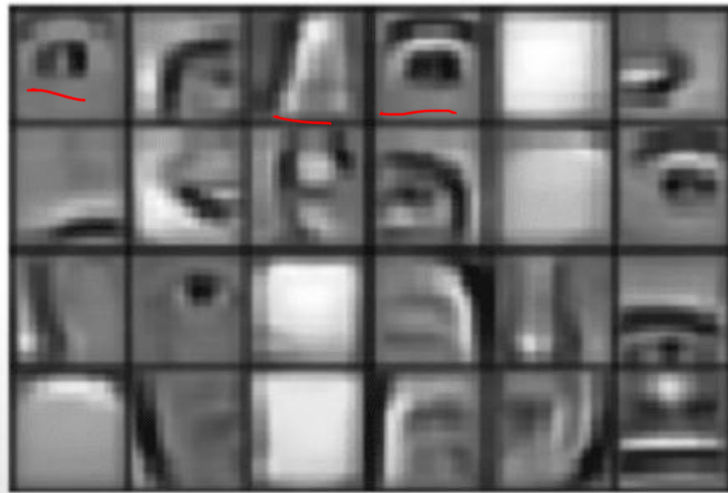
Deep Learning

Low level features



Edges, dark spots

Mid level features



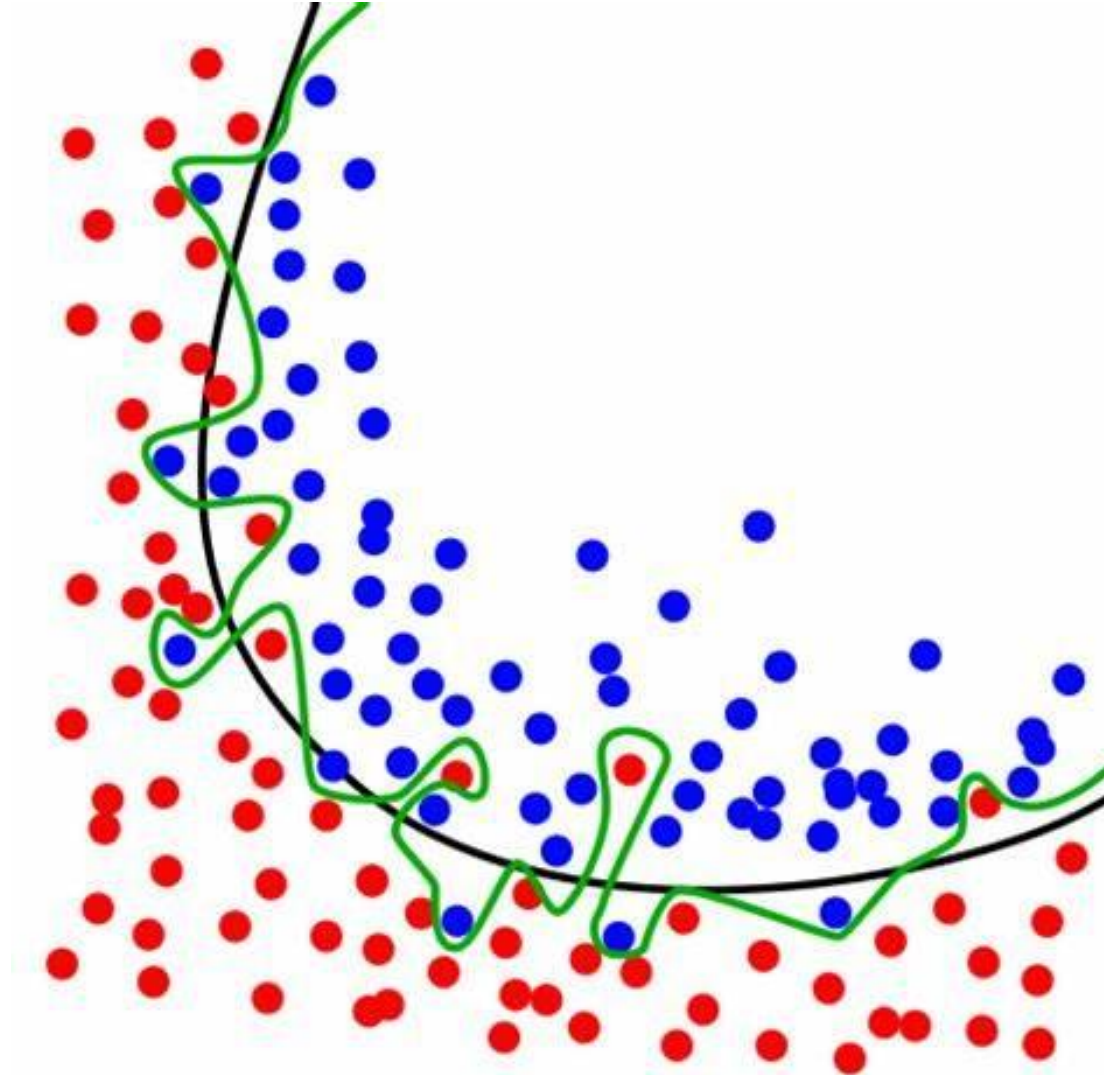
Eyes, ears, nose

High level features

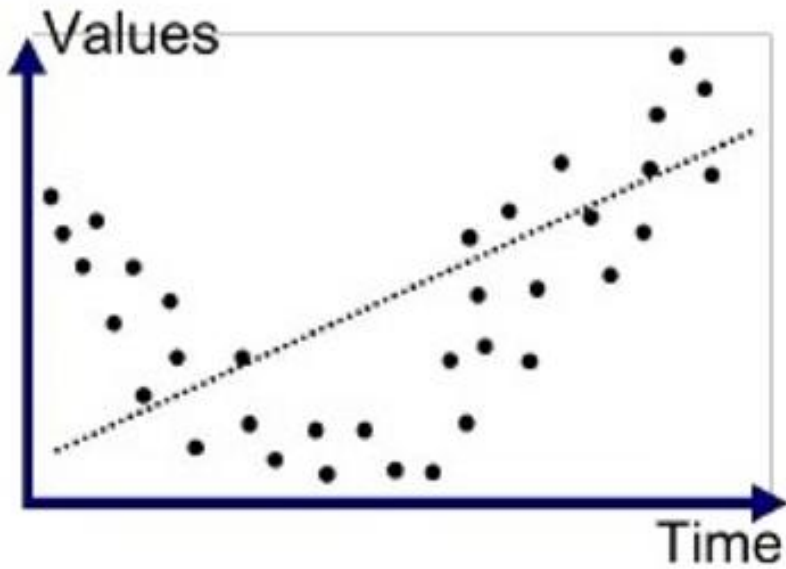


Facial structure

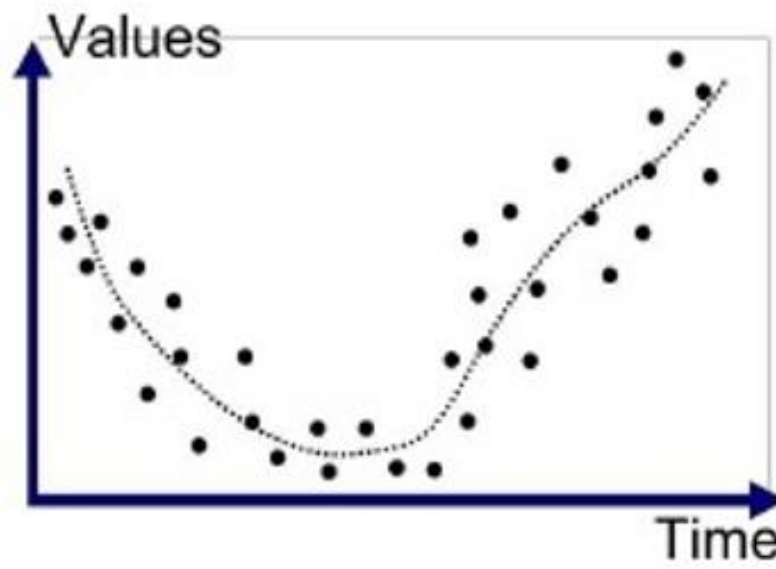
Overfitting e Underfitting



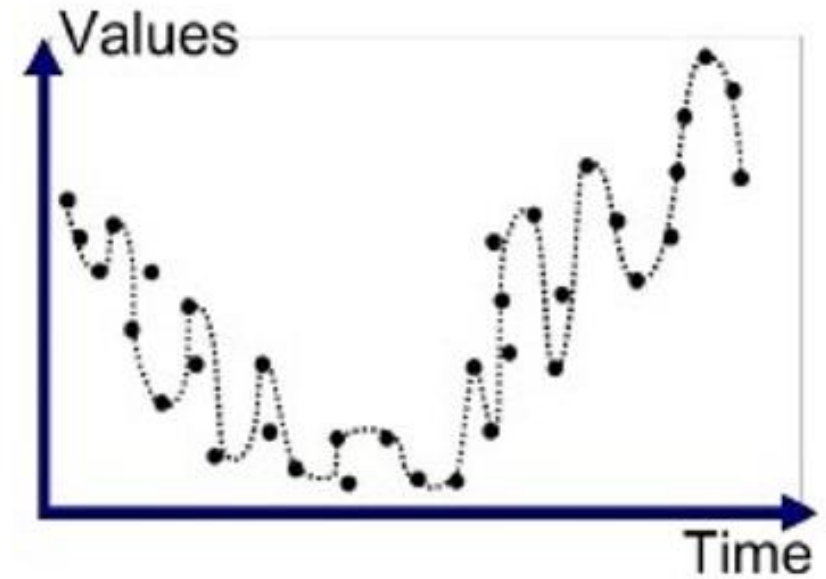
Overfitting e Underfitting



Underfitted

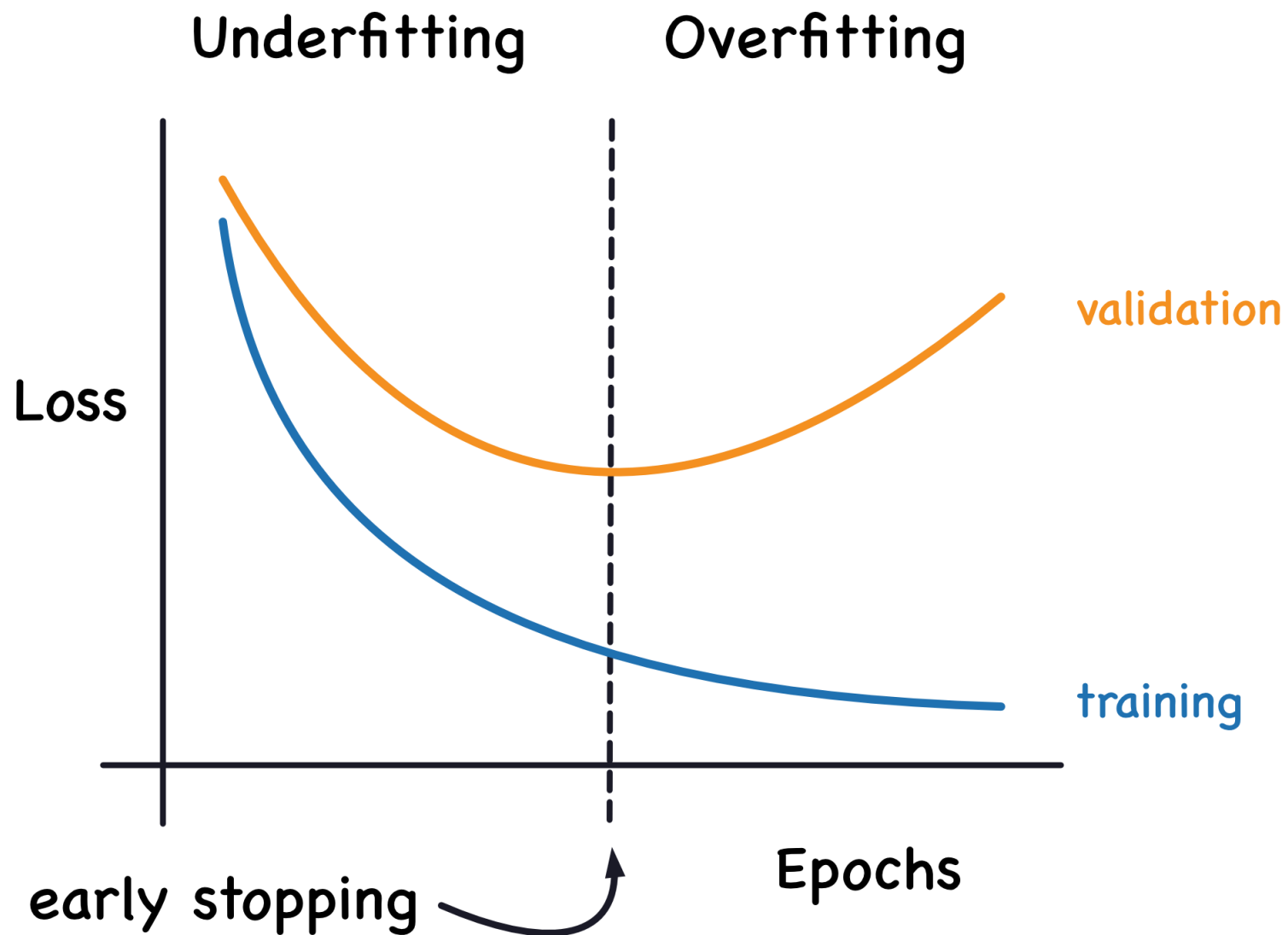


Good Fit/Robust



Overfitted

Overfitting e Underfitting



Overfitting

